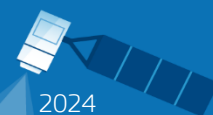


Earth Observation Strategic Research and Innovation Agenda

Enabling Copernicus evolution

Rixen M., Dowell M., Immler F., Dusart J., Kristopaitis E.



This document is a publication by the Joint Research Centre (JRC), the European Commission's science and knowledge service. It aims to provide evidence-based scientific support to the European policymaking process. The contents of this publication do not necessarily reflect the position or opinion of the European Commission. Neither the European Commission nor any person acting on behalf of the Commission is responsible for the use that might be made of this publication. For information on the methodology and quality underlying the data used in this publication for which the source is neither Eurostat nor other Commission services, users should contact the referenced source. The designations employed and the presentation of material on the maps do not imply the expression of any opinion whatsoever on the part of the European Union concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries.

Contact information

Name: Mark DOWELL

Email: mark.dowell@ec.europa.eu

EU Science Hub

<https://joint-research-centre.ec.europa.eu>

JRC136730

EUR 31903 EN

PDF ISBN 978-92-68-14333-9 ISSN 1831-9424 doi:10.2760/18985

KJ-NA-31-903-EN-N

Luxembourg: Publications Office of the European Union, 2024

© European Union, 2024



The reuse policy of the European Commission documents is implemented by the Commission Decision 2011/833/EU of 12 December 2011 on the reuse of Commission documents (OJ L 330, 14.12.2011, p. 39). Unless otherwise noted, the reuse of this document is authorised under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence (<https://creativecommons.org/licenses/by/4.0/>). This means that reuse is allowed provided appropriate credit is given and any changes are indicated.

For any use or reproduction of photos or other material that is not owned by the European Union permission must be sought directly from the copyright holders.

- Cover page illustration, © European Union, Copernicus Sentinel-2 imagery, from the Sentinel-2 Global Mosaic (<https://s2gm.land.copernicus.eu/>)

How to cite this report: European Commission, Joint Research Centre, Rixen, M., Dowell, M., Immler, F., Dusart, J. and Kristopaitis, E., *Earth Observation Strategic Research and Innovation Agenda*, Publications Office of the European Union, Luxembourg, 2024, <https://data.europa.eu/doi/10.2760/18985>, JRC136730.

Contents

Abstract.....	3
Executive summary.....	4
Acknowledgements.....	4
Introduction	5
1.1 Scope of the Strategic Research and Innovation Agenda.....	5
1.2 Copernicus and the EU Space Programme	5
1.3 EU Strategy for Space R&I.....	6
1.4 Horizon Europe and Earth Observation	6
1.5 KCEO and user engagement.....	8
1.6 Consultation Process and update plan	9
2 Context.....	10
2.1 Research for a user-driven programme.....	10
2.2 Core user uptake and emerging EU Policy needs	10
2.3 Global Agenda and GEO	13
2.4 End-to-end value chain and High-level SWOT analysis.....	15
2.5 Copernicus evolution	16
3 Service priorities	18
3.1 Copernicus Climate Change Service (C3S)	18
3.2 Copernicus Atmosphere Monitoring Service (CAMS)	20
3.3 Copernicus Emergency Management Service (CEMS).....	22
3.4 Copernicus Land Monitoring Service (CLMS).....	25
3.5 Copernicus Marine Environment Monitoring Service (CMEMS)	27
3.6 Copernicus Security Service (CSS).....	30
4 Upstream and downstream elements	38
4.1 Space and ground segments - ESA.....	38
4.2 Space and ground segments - EUMETSAT	44
4.3 In-situ networks– - EEA	47
4.4 Market uptake– - EUSPA.....	49
5 Horizontal elements.....	52
5.1 Thematic Hubs	52
5.2 Computing and data infrastructures.....	53

5.3	Digital Twins	54
5.4	Artificial Intelligence/Machine Learning	55
5.5	Synergies and partnerships at European and Global level	56
6	Measure of success and implementation recommendations	58
7	Conclusions	60
	References	61
	List of figures	62

Abstract

This first Strategic Research and Innovation Agenda (SRIA) for Earth Observation sets out a comprehensive framework for Copernicus' research and technological development. It outlines key priorities for the Horizon Europe Work Programmes for 2025-2027 and beyond into the next Multi-annual Financial Framework, ensuring that Copernicus and its products and services remain state-of-the-art.

Copernicus, a key component of the EU Space Programme, provides Earth observation data through satellites, contributing missions, and in-situ data sources. This wealth of information is transformed into actionable services across six thematic areas: land, ocean, atmosphere, climate change, emergency, and security. The SRIA aims to ensure the continuity and evolution of these services.

The SRIA outlines how Earth Observation (EO) is integral to the EU Policy goals, such as in the implementation of the European Green Deal. The SRIA will contribute to guiding Copernicus' evolution, emphasizing agile, resilient data streams, digital transformation, and expanded services. Enhancing policy uptake and downstream adoption is crucial, ensuring Copernicus meets evolving user needs and drives economic growth, societal well-being, and security in Europe. The report emphasizes the need for comprehensive support for Copernicus' services, ensuring they stay relevant and beneficial for European and global policy goals. The SRIA addresses the individual R&I needs of each service and their components.

The SRIA also addresses horizontal elements and priorities to create synergies and efficiencies across services. Key initiatives include Thematic Hubs, which centralize access, enhance services, and target specific cross-cutting applications. Computing and data infrastructures, Digital Twins, and AI/ML also play pivotal and growing roles in advancing Earth observation. Coordination across entities, EU policies, and global organizations fosters seamless integration, maximizing Copernicus' impact. The strategy emphasizes the evolution of services and technologies, particularly through research, data integration, and user-centric developments. Collaborative efforts with European and global partners further strengthen Copernicus, aligning it with broader policy goals and ensuring effective implementation across diverse sectors and communities.

The SRIA concludes by emphasizing three dimensions that need to be addressed in its implementing. These are the Earth observation research and innovation activities themselves, the delivery and evolution of sustained Earth Observation services and the underlying policy needs for data and information derived from Earth observation. To monitor the successful implementation of the strategy a key focus should be on the interfaces and connections between the different dimensions identified. This then allows to identify specific metrics that can be followed.

Executive summary

This report outlines the new Strategic Research and Innovation Agenda (SRIA) for Earth observations enabling the Copernicus evolution for the remaining part of the Multiannual Financial Framework (MFF) and beyond. It represents a major update of the roadmap developed back in 2020 for the current MFF and was justified by growing user requirements, new technologies, evolving policies at EU and global level and a changing global geopolitical context.

The Copernicus evolution for the period 2021-2028 focuses on continued and more agile observations, service uplifting to take advantage of the digital transformation (artificial intelligence and machine learning, cloud and edge computing, *etc.*), more federated data infrastructures to enable data analytics, and promoting downstream market uptake of Copernicus products.

At this mid-term juncture, users require increased resilience and secure earth observations, enhanced knowledge provision, interactivity and decision-making tools, interoperability and AI-readiness, which can enable mass adoption and on-demand offer.

The SRIA provides a longer-term perspective for Research and Innovation to inform the Copernicus evolution but also to justify and leverage investments in Earth observation. It complements or supplements other assessments such as those developed by Entrusted Entities and agencies to look at the broader potential of Earth observation beyond the strict EU Space Programme and its Copernicus component standpoint. Whilst Copernicus provides short-to-medium term research support at pre-operational level, the scope of the SRIA is to help shaping the long-term evolution of Copernicus and Earth observation across the European Union Research Programme. In practice, the document will serve as a reference to plan and develop associated research calls and topics over the years to come.

Acknowledgements

The development of the report was jointly coordinated by DG DEFIS, JRC and DG RTD, under the umbrella of the Knowledge Centre on Earth Observation (KCEO) to mainstream research and innovation across Horizon Europe. It builds on intense consultations with and contributions from the Copernicus Entrusted Entities (EEA, ECMWF, EMSA, Frontex, JRC, Mercator Ocean International, SatCen) and Agencies implementing Copernicus (ESA, EUMETSAT, EUSPA), numerous EU Directorate Generals and Executive Agencies, and Member State delegates to the Copernicus User Forum and Committee.

The authors are indebted for all these extremely valuable contributions which are hereby deeply acknowledged.

Finally, the authors are also grateful for the contribution in editing and typesetting the report provided by Camilla Bertoglio and Anna Kemppinen.

Introduction

1.1 Scope of the Strategic Research and Innovation Agenda

The scope of the present document is to summarize and outline the Strategic Research and Innovation Agenda (SRIA) for Copernicus and Earth observations⁽¹⁾, the aim of which is to serve as a framework and reference for calls under the upcoming Horizon Europe⁽²⁾ Work Programmes for period 2025-2027 and initial preparation for the subsequent EU Research and Innovation framework programme 2028-2035. It integrates the Strategic Research Agenda for the Copernicus Security Service (SRA-CSS) and represents an update of the Copernicus Service Evolution roadmap developed back in 2020.

It addresses requirements of the Union Space Programme⁽³⁾ to support the Union's research and technological development activities carried out under Regulation (EU) 2021/695 [Horizon Europe].

1.2 Copernicus and the EU Space Programme

Copernicus is the Earth observation component of the EU Space programme, looking at our planet and its environment to benefit all European citizens, their wellbeing and security. The Copernicus backbone relies on a set of dedicated satellites (the current Sentinel family and upcoming Sentinel Expansion and Sentinel Next Generation missions), contributing missions (public and commercial satellites), and in-situ (non-space) data.

Copernicus services process, analyse, transform, and exploit the wealth of satellite and non-satellite ('in-situ') data into numerical models, products and applications towards timely and actionable information to help service providers, public authorities, and other international organisations improve European citizens' quality of life and beyond. These value-adding activities are streamlined through six thematic core services covering the land, ocean, atmosphere, climate change, emergency and security domains, the scope of which is defined in the Space Programme Regulation. The information services, as well as the data from which they are derived, are accessible on a full, free, and open basis by anyone.

The technological innovation, maturity, operational commitment, as well as the free and open data policy of Copernicus have positioned the EU and Copernicus as major players in today's space-based activities.

⁽¹⁾ Earth observation shall be understood here across the entire space upstream and downstream value chain of geospatial data on the environmental state of the Earth, including from in-situ instruments and other non-space based sensors, numerical models, as well as product, applications and services.

⁽²⁾ [Regulation \(EU\) 2021/695 of the European Parliament and of the Council of 28 April 2021 establishing Horizon Europe – the Framework Programme for Research and Innovation, laying down its rules for participation and dissemination, and repealing Regulations \(EU\) No 1290/2013 and \(EU\) No 1291/2013 \(Text with EEA relevance\)](#)
[Text with EEA relevance](#)

⁽³⁾ [Regulation \(EU\) 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme and repealing Regulations \(EU\) No 912/2010, \(EU\) No 1285/2013 and \(EU\) No 377/2014 and Decision No 541/2014/EU](#)

1.3 EU Strategy for Space R&I

During the next decade, the EU will consolidate its position as a global space actor able to access space, to act in space, to communicate and navigate in space and on Earth, and to monitor space and the Earth for the well-being and security of its Member States and citizens.

To achieve this, the European Commission is working towards a future space ecosystem that is competitive, sustainable, resilient, and future-oriented. It should be underpinned by modern space flagships (Galileo, EGNOS, Copernicus, GOVSATCOM, SSA, and IRIS2) in-line with user requirements and demands.

Space R&I has a considerable impact, not only on the growth of the space sector itself, but also on benefits to society in general. The spin-offs and cross fertilisation of R&I activities support the innovativeness of both the upstream and the downstream sectors. From a broader societal standpoint way beyond the space sector, there is substantial and quantitative evidence that space related research and development yields huge societal, economic, and environmental benefits for European citizens and end-users, estimated at 108.8 B EUR for the timeframe of 2017 – 2035³ (a tenfold yield compared to the investment).

An overall *EU Strategy for Space R&I* is needed taking a holistic approach towards an ambitious and comprehensive plan to support its Space Programme Flagships, and to actively drive the EU's strategic autonomy, resilience and freedom of action in space and on Earth.

Copernicus, in coherence with national and other inter-governmental satellite programmes, shall evolve in this context to provide state-of-the-art uninterrupted, high-quality, up-to-date and, free and open or where appropriate, secure, Earth observation data, products, applications and services to sustainably manage the environment we live in, in support of EU's political agenda.

1.4 Horizon Europe and Earth Observation

The vision of the EU Research and Innovation Programme Horizon Europe is, inter alia, to

- tackle climate change,
- achieve the UN's Sustainable Development Goals,
- boost the EU's resilience, competitiveness, and growth,
- facilitate collaboration and strengthen the impact of research and innovation in developing, supporting, and implementing EU policies while tackling global challenges.

Earth observation (EO) plays an important role in addressing these goals by providing evidence and scientific understanding of the issues at stake and monitoring tools for environmental policy implementation. This includes the European Green Deal, as well as support for smart decisions to all sorts of end-users. Therefore, EO plays a role in all parts of Horizon Europe and related activities can be found in all 3 Pillars: Excellent science (Pillar I), Global challenges and European industrial competitiveness (Pillar II) and Innovative Europe (Pillar III), and in particular in all the global challenges of Pillar II.

Building upon the first Strategic Plan covering the years 2021-2024, the second Horizon Europe Strategic Plan sets out the key strategic orientations for research and innovation investments over the period 2025-2027, acting as a compass to stay on course in the green and digital transition, responding to new challenges and taking into account the lessons learned since the launch of Horizon Europe and from Horizon 2020, which are captured in the last section of this document in the form of implementation recommendations.

The Strategic Plan which constitutes the basis for the drafting of the next Horizon Europe Work Programmes builds on the input provided via public consultations, and an in-depth Horizon Europe strategic plan 2025-2027 analysis. This analysis identifies gaps in terms of R&D in all clusters and intervention areas.

Earth observations, while considered in all pillars of Horizon Europe, is formally addressed under Pillar II in the **Cluster 4 on “Digital, Industry and Space”**. The bullet points below provide examples of potential gaps that could be considered in the next work programmes under Pillar II:

- Under **Cluster 2 on “Culture, Creativity and Inclusive society”**, in the intervention areas on European cultural heritage and the cultural and creative industries and on social and economic transformations, *EO could play a role in developing innovative research and innovation actions in combination with Social Sciences and Humanities to support workers’ wellbeing, climate change resilient work environments, effective inclusion and vocational training for a fair twin transition towards climate neutrality.*
- Under **Cluster 3 on “Civil security for society”**, in the intervention area on Disaster-resilient societies, EO could play a role in helping better addressing *transport disasters, post-disaster recovery, economic of disasters and nature-based solutions for prevention.* There is a well an expressed need to improve *ways of dealing with climate-related disaster events and the increased tendency for cascading and compound events.*
- Under **Cluster 4 on “Digital, Industry and Space”**, in the intervention area on Space, including Earth Observation, there is an expressed need to *foster the EU’s space sector competitiveness, focusing on areas not yet addressed* and the need to *continue to evolve the Copernicus services.* The gap analysis calls as well to the *development of applications relying on the combined use of Galileo and Copernicus.* In the intervention area on Digital, there are explicit R&I needs for AI and Digital Twins (see dedicated sections).
- Under **Cluster 5 on “Climate, energy and mobility”**, in the intervention area on Climate science and solutions, a potential gap is listed on *the need for better understanding and monitoring of ocean & terrestrial carbon fluxes, and improvements of the methods for their monitoring on different spatial and temporal scales.* Another identified gap calls for *the need for better understanding, prediction and attribution of extreme climate events – with a focus on the development of effective climate services and early warning systems.*
- Under **Cluster 6 on “Food, bioeconomy, natural resources, agriculture and environment”**, in the intervention area on Environmental observation, the gaps identified focus on environmental observations and include EO in the Polar Regions; ocean observations; improvement of EU governance on EO, international coordination, development of standards and good practices, and interoperability and harmonisation of data management and data technologies; Support for Europe’s part of GEO initiative, and other initiatives of GEO, with focus on Africa and South America; EO to understand and predict the climate, and support climate mitigation and adaptation, including loss and damage; Monitoring and assessment of biodiversity; Integration of biodiversity data across domains and methods; Zero pollution, covering pollution in oceans and water (including groundwater, underwater noise), air and soil pollution, microplastics and nanoplastics, nutrients and pollutants of emerging concern; Tools to optimize observing systems; Improvement of the monitoring of air quality; Agricultural monitoring; EO services to support food security; EO for resilient, climate-neutral and healthy cities and communities.

European and national entities (e.g. ESA, EUMETSAT, national space programmes in EU Member States) are strengthening the European space and ground capacities to tackle the political priorities of the Commission (Green Deal, Digital Agenda, ...). In the area of the monitoring of weather and climate from space, new and innovative generations of satellite systems (MTG, EPS-SG, EPS-Aeolus, EPS-Sterna, etc.) will play a key role in enabling some Copernicus services and provide opportunities for R&I in several Clusters (Civil security, climate, environment).

Under Pillar I, bottom-up mechanisms such as the European Research Council (ERC) and Marie Skłodowska-Curie Actions (MSCA) provide opportunities to leverage breakthrough not identified in top-down planning. Some Research Infrastructures under Pillar I, such as ICOS and ACTRIS, also become essential components of some of the services, thereby challenging their ‘research’ status.

Under Pillar III, the European Innovation Council (EIC) supports game-changing innovations and high-risk/high pay-off ideas of Earth Observation SMEs & start-ups in both the downstream and upstream ecosystems, enabling them to develop market driven innovations, helping them become competitive and leading players in the fast-evolving global space industry through its Pathfinder (TRL1-4), Transition (TRL4-6) and Accelerator programs (TRL6-9). The work program encompasses both a “bottom-up” aspect through its “open” element and “top-down” one through the “challenge” instrument. The “challenge” topics are proposed by the EIC Programme Managers. Historically, EO projects have entered the EIC space portfolio through the EIC Accelerator open program.

The [Horizon Europe Missions](#) aim at addressing important horizontal challenges (adaptation to climate change, healthy soils, ocean and water restoration, climate neutral and smart cities, and cancer) in Horizon Europe and other Union instruments. Examples of activities of the Missions of direct relevance to the EO community include monitoring, reporting and verification tools for soil carbon removals, Digital Twins of the Ocean (DTO) or better understanding of climate risks and associated adaptation measures. The recently adopted Communication on **“EU Missions two years on: assessment of progress and way forward”**⁽⁴⁾ reaffirms the commitment of the Commission to maintain the EU Missions as flagship activities.

In line with the new strategic plan 2025-2027 adopted on 20 March 2024, the first Horizon Europe work programme 2025 will start to be prepared in Q2-Q3 of 2024, with an expected adoption in late Q1/early Q2 2025. The timeline of the last years of Horizon Europe is still to be defined, while the work on the successor of Horizon Europe has started, looking first at lessons learned from previous work programmes and overarching priorities that will have to be confirmed by the new College after the European elections.

1.5 KCEO and user engagement

The Knowledge Centre on Earth Observation (KCEO) launched in 2021, aims to strengthen the internal (i.e., pan-Commission) uptake and coordination of Copernicus/EO support to policies. The need for such a dedicated internal uptake/coordination tool has become clear in recent years due to the fast-growing number of policy areas that are interested in making use of EO information to support their policies. It is also relevant to take a broader view which encompass Copernicus but also national or intergovernmental relevant EO satellites programmes, such as the European meteorological programmes.

⁽⁴⁾ [https://ec.europa.eu/transparency/documents-register/detail?ref=COM\(2023\)457&lang=en](https://ec.europa.eu/transparency/documents-register/detail?ref=COM(2023)457&lang=en)

The Knowledge Centre aims to act as an efficient coordination mechanism inside the Commission to maximise the use of products and information from Copernicus, and Earth observation in EU policy-making and implementation.

The work of the KCEO will focus on two main pillars:

- increase policy uptake of Copernicus and other EO products and services, in partnership with user DGs, and ensure the increased use of Copernicus and other EO programmes in addressing policy coherence. This is addressed by identifying policy priorities and through “deep dive” assessments of the needs and requirements;
- undertake a continuous scan for prioritising research and development investments, thus ensuring that the Earth observation capacities in Europe remains state-of-the-art and responsive to new needs for data and services to support the relevant policies. Efforts are also made on mainstreaming Earth Observation R&I across all areas of the Framework Programme, and beyond.

While the KCEO addresses primarily the Commission services, its ambitions are as well to serve the EO needs and requirements of the EU and its Member States to support the monitoring, implementation and reporting duties of international environmental agreements the European Union has ratified. As such, it offered a perfect home to develop the SRIA.

1.6 Consultation Process and update plan

In line with the Horizon Europe Regulation, the Copernicus programme has developed its own stakeholder consultation to identify its own R&D needs and priorities, involving Entrusted Entities (in charge of the Copernicus core services), the user communities and the Copernicus Committee and Copernicus User Forum. The development of the SRIA was conducted under the KCEO umbrella jointly by DG DEFIS, DG JRC and DG RTD in three main phases. The first one involved three workshops held during the first half of 2023 to consult respectively:

- Entrusted Entities and implementing partners (*e.g.* ECMWF, MOi, ESA, EUMETSAT, EEA, JRC, Frontex, EMSA, SatCen) on lessons learned from Copernicus 2.0;
- EU Directorate Generals on existing and emerging policy requirements, and Agencies (such as HaDEA, CINEA, REA and EUSPA) on implementation lessons;
- All stakeholders, including Copernicus User Forum and Copernicus Committee delegates, on horizontal issues (Thematic Hubs; Artificial Intelligence/Machine Learning; Data infrastructures, computing and cloud; International cooperation and partnerships).

The second step, held during the second half of 2023 involved content development of the present document informed by the outcomes of the above meetings as well as specific inputs received from entities and agencies. The last step concerned the endorsement of the present document by the Copernicus governance.

The document should hence serve as a reference for calls issued for the WP2025-2027 as well as setting broad directions for the next MFF for which a further update of the SRIA would be prepared before 2028.

2 Context

2.1 Research for a user-driven programme

Copernicus as a user-driven programme, was established to support EU and Member States policy requirements. The Space Programme Regulation makes the distinction between “core users” and “other users”. The Core Users include Union institutions and bodies and European, national, or regional public bodies in the Union or Copernicus Participating States uptake of Earth Observation. The other users reflect all other users, including research and education organisations, commercial and private bodies, charities, non-governmental organisations and international organisations.

A fundamental aspect of the Regulation is that the core user requirements should drive the evolution of the programme. As new/additional user requirements become available from core users, there is a need to analyse the implications for the evolution of the programme. This includes making an assessment of gaps and opportunities with respect to the existing operational programme elements and components, which need to be addressed by additional research and innovation to advance the state-of-the-art.

An important consideration is to understand the cascade and feedback along the value chain. As new needs and requirements are expressed, we should assess the fitness for purpose of existing products and services to address these as follows:

- firstly, an assessment on the implications of requirements on the evolution of Core Services;
- secondly, if an identified gap within the service is dependent on missing products provided from the ground segment, this may have additional implications for the evolution of products and the ground segment;
- finally, if the gap still remains at the level of the ground segment, implying missing observations, this may have longer-term consequences for the evolution of the space component of the programme.

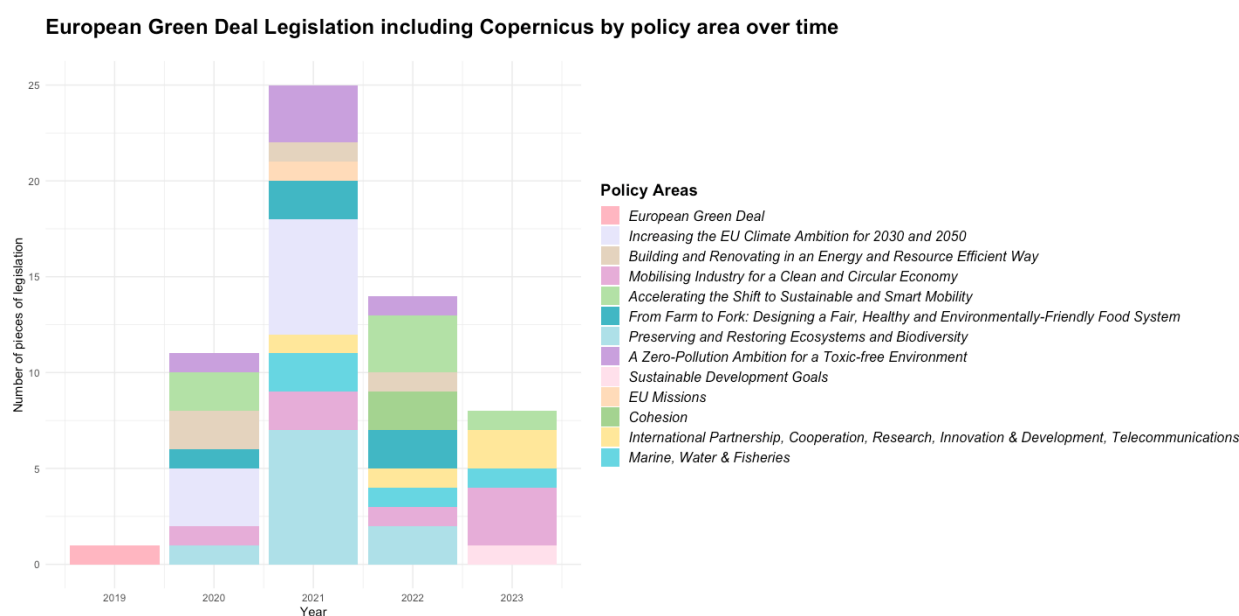
Each step in this cascade process may also require additional research and innovation to bridge the gap between the different stages of the value chain, which can be addressed both through the Framework Programme and potentially from institutional research programmes of Copernicus partners (*e.g.* ESA, EUMETSAT). The user’s requirement process undertaken by research and operational space agencies is a critical scouting function for future application areas. In undertaking a systematic approach to assessing the fitness-for-purpose gap assessment and cascading implications for the evolution throughout the value chain will provide a robust framework to ensure the evolution of the programme and advancing the state-of-the-art of the EU Earth observation capabilities traceable back to emerging user and policy needs of core users. The user-driven pull of the Space Programme should also closely monitor new science and technology developments such as NewSpace capabilities, artificial intelligence, *etc.*, to stimulate broader and state-of-the-art services.

2.2 Core user uptake and emerging EU Policy needs

The current MFF has seen a fundamental change in the recognition of the value of Earth observation and Copernicus in supporting policy. Whilst prior to this MFF there were less than 10 explicit references to Copernicus in legislation in total, in the first 3 years of the current MFF 2021-2027 we have seen already

30+ references to Copernicus. This has been especially true, but not restricted to new legislation in the context of the European Green Deal (see Figure below). The references are across EGD legislation including Regulations, Directives, Communications and other formal documents. The breadth of policy areas the references address include Circular Economy, Finance Energy, Industry, Marine, Zero Pollution, Climate, Biodiversity, Forestry, Environment, Sustainable Development Goals, Missions, Agriculture and Cohesion. Some examples include the European Climate Law and the Zero Emissions Strategy, the Nature Restoration Law, the Strategy on adaptation to climate change, the EU Soil Strategy for 2030, Communication on Sustainable Carbon Cycles, the EU Forest Strategy for 2030, the Marine Strategy Framework Directive, Stepping up EU Action to Protect and Restore the World's Forests, or the European Methane Action Plan.

Figure 1. European Green Deal Legislation including Copernicus by policy area over time



In general terms these references relate to providing underpinning data for framing the policy development, defining priorities and setting targets, supporting the implementation through dedicated information services, and for regular and transparent progress tracking and performance or compliance assessments. A notable change which has occurred in the last years is a transition from technological solutions being pushed on policy DGs to them making proactive demands for support with specific products and services in support of their policy files. This has been seen very clearly in the initial assessments being made through the KCEO for policy domains such as Biodiversity and Urban Climate Adaptation as well as *ad hoc* requests in a number of areas.

To further identify the evolving needs for Copernicus in the future, one can look at the existing and planned legislation and the policy milestones (e.g. in European Climate Law) that are foreseen therein for the current MFF, the next MFF and beyond – thus having implications for service, products and satellite constellation evolution well into the next decade.

In terms of gaps with respect to policies being implemented and the evolution of Copernicus, several areas are becoming apparent through discussions with the DGs, and the evolution of Copernicus for different applications would be required to ensure increased uptake for EU policy:

- More focus on land use and land use change aspects at high granularity levels, such as detailed urban areas and various aspects of the “green transition” (sustainable land use, degradation).
- Biodiversity is a key evolving field with obvious new Copernicus contributions; in particular when linking with ground-based datasets.
- Climate change related monitoring will have to put more focus on assessing and supporting climate actions, including for Measuring, Reporting and Verification (MRV), and for efficient use of climate finance.
- Linking statistical surveys/assessments and EO data is important for monitoring activities by Member States and leverage Copernicus opportunities.
- Expanding traditional monitoring to new areas (i.e., cultural heritage areas, social wellbeing, and inclusion, focus on different regions worldwide – Africa, Latin America and Caribbean, Arctic, Antarctic).

A crosscutting aspect that is emerging and could have serious implication for how Copernicus is used in policy uptake involves the slow transition from a pure monitoring tool to a possible compliance tool of EU policies. Copernicus is already a mandatory tool for some (i.e., agriculture) and further examples of use of EO-data for compliance monitoring. In this context, quality control, transparency and traceability will become increasingly critical. These include Common Agricultural Policy – accurate, annual delineation and area measurements of ‘all’ agricultural parcels. Specific policy areas where the potential enforcement aspects from Copernicus will become increasingly important include Climate policy including spatially explicit tracking of climate actions - GHG sinks and sources; degradation of the protected habitats and species in Natura 2000 sites, monitoring EU just transition aspects such as energy and transport poverty, and illegal fishing, mining, logging, *etc.*, worldwide.

Some of the policy needs, such as those expressed in the latest Staff Working Document⁽⁵⁾ on the subject have general implications. There is a recurrent need for Copernicus to become spatially more detailed. This implies the detection, characterization and tracking of small objects such as buildings (size, location, roofs, greening, heat loss), traffic (volumes, types), small landscape features (i.e., in agricultural landscapes), pollutions sources, animals (movements/behaviour, counts, GHG emissions), disasters/conflicts and impacts (incl. human displacements).

Aside the policy DGs have expressed additional important suggestions related to the operational, long-term nature of Copernicus, including:

- Importance of providing services in long-term, consistent manner, based on long consistent timeseries of observations.
- Need for higher frequency updates and lower latency of the delivery products (including near-real time forecasts).
- Need of very high spatial resolution data from observations and models.
- Need for more comprehensive thematic characterizations requiring new type of data such as hyperspectral (i.e., crop types, biodiversity, pests), thermal (i.e., animals, buildings/heat loss),

⁽⁵⁾ Expression of User Needs for the Copernicus Programme, Staff Working Document (2019), 394 final, 139pp

LiDAR (i.e., building/tree height changes, vegetation structure), RADAR (i.e., cloud-free capabilities to increase temporal detail) which may be addressed not only by satellite data but also additional monitoring networks (e.g. airborne, ground-based).

- Need for additional development of Copernicus products for habitat mapping, integrating data across sensors and environmental parameters.
- System resilience along the entire Copernicus value chain is key to permit un-interrupted operational monitoring to ensure business continuity if Copernicus is to be used in official reporting/statistics.
- Consistency of monitoring across domains (i.e., atmospheric, ocean, and land monitoring for climate actions), e.g. on carbon, energy and water cycles.
- Need for analysis-ready and user-friendly processed Copernicus data, products and last mile policy applications to be delivered in ways for easy uptake by DGs.
- Interest in global monitoring with Africa is a key priority outside Europe.

It is apparent that there is still often a gap between the needs of the policy DGs for tailored products and services and those that can be delivered through the Core Services as defined in Copernicus. Mechanisms need to be found to implement these *last-mile* policy applications if the uptake of Copernicus in EU Policy is to reach its full potential. As an example, there is certainly potential to streamline EO applications that involve spatial environments of cross-policy relevance where the cost of very high-resolution imagery could be shared between different policy DGs.

Many of these emerging requirements are already well recognised and are addressed both by existing research projects and in the statements on upcoming priorities by Entrusted Entities in Chapter 3 of this document. The remaining elements, many of which would address the needs for multiple policies and promote policy coherence (based on objective data) should be included in future priorities.

User uptake activities are meant to grow the Copernicus user database by reaching out to different user communities, from the EU and Member States institutional users to the broader global community, from the public sector to the private sector. A central focus of Copernicus is on core user uptake. This represents a joint effort across many stakeholders (e.g. DG DEFIS, Entrusted Entities and Agencies, KCEO), and involves instruments such as the National Collaboration Programmes, Thematic Hubs, etc. We refer to the relevant service sections for policy and core user uptake in particular.

2.3 Global Agenda and GEO

Full, free and open access to Earth observation data, information and knowledge is crucial for humanity as it faces unprecedented global social, economic, environmental, emergency and security challenges.

The United Nations (UN) established bodies, multilateral agreements and frameworks aiming at addressing the above issues, such as the Sustainable Development Goals (SDG), the Intergovernmental Panel on Climate Change (IPCC), the United Nations Framework Convention on Climate Change (UNFCCC), including national commitments under the Paris Agreement of 2015, the Sendai Framework for Disaster Risk Reduction, the WMO and its Global Framework for Climate Services, the Early Warning for all initiative, the UNESCO-IOC, the UN Decade of Ocean Science, the UN Ocean Conference, and other conventions such as the UN Convention on Biological Diversity, the UN Convention on the Law of the Sea the UN Convention to Combat Desertification, the UN Convention on Long-Range Transboundary Pollution, the Montreal Protocol on Substances that deplete the Ozone Layer, etc.

Those multilateral frameworks are greatly benefitting from Copernicus and other European observing systems and infrastructures (like ICOS, EuroArgo, *etc.*). The European EO sector can strongly support EU commitments towards this global agenda.

In this context, it is important to note the need to provide products and services in a consistent manner so that they can be used both in EU Policy directly as well as in the global provision of data for our commitments to these international Conventions and Treaties. For example, the use of Copernicus both in support of the EU Biodiversity Strategy and in the context of the indicators for the Global Biodiversity Framework.

The Group on Earth Observation (GEO), a partnership of more than 100 national governments and more than 100 Participating Organizations, envisions a world where trusted Earth intelligence is universally accessible and empowers society to achieve a sustainable future. The term Earth Intelligence refers to integrated Earth and social science derived knowledge and insights that inform strategic decisions⁽⁶⁾. GEO advocates open sharing of EO data and knowledge for effective measures addressing the global environmental challenges.

As one of the four co-chairs of GEO, the European Commission is engaged in the governance of GEO and contributes mainly with the EU's R&I framework programmes to GEO work programme activities, i.e. the GEO flagships, initiatives and pilot initiatives. References to GEO are added in the Horizon Europe Work Programme to ensure the effectiveness of this contribution. Moreover, DG JRC leads the initiatives "GEO Human Planet" and "Global Wildfire Information System (GWIS)". Initiatives like these drive the international engagement and support the implementation of the multilateral agreements listed above.

Copernicus, with its free and open data policy, is a major European contribution to GEO, which offers a unique platform for international valorisation of the EU's investment in EO. EuroGEO, the European GEO component, brings together environmental and Earth observation researchers and innovators in Europe, and creates impact and visibility on the global landscape. Those linkages between Copernicus and EuroGEO should be further strengthened, for example via the entrusted entities, especially on R&D. The EuroGEO community reflects the broad range of stakeholders involved in support to Europe's contributions. This includes the research community, member state agencies, the private sector, European institutions and Entrusted entities.

EuroGEO implements a user-driven research and innovation agenda and develops EO applications that support user communities addressing EU policies like the European Green Deal and GEO priorities. Supported by Horizon Europe, EuroGEO fosters demonstration, incubation, up-scaling, and replication. EuroGEO also coordinates the European contribution to the GEO global initiatives and priorities. EuroGEO thus supports the valorisation of Copernicus outputs by delivering user-tailored services in Europe, and also on an international scale, through the GEO initiatives.

⁽⁶⁾ <https://earthobservations.org/storage/app/media/documents/Key-Documents/GEO%20Post%202025%20Strategy%20Full%20Document.pdf>

2.4 End-to-end value chain and High-level SWOT analysis

Copernicus is the Earth observation component of the EU space programme and is also responsible for the full value chain of the system with its core services and its upstream and downstream components, including space, in-situ and ground processing segments, as well as applications and services value adding activities to serve public/institutional users as well as the broader market uptake. A recent Expert Group report⁽⁷⁾ pointed to the need to enhance system resilience and non-dependence, to address policy requirements, to foster the commercial sector contribution, and to upgrade the space and ground infrastructures. The strategic SWOT analysis on Copernicus flagged the following main points:

- *Strength*: since it moved into operations, Copernicus became a world top provider of state-of-the-art and traceable EO big, continuous, reliable, free and open data and products, applications and services, exploited by a wide use base (hundreds of thousands), part of which even relying critically on Copernicus for their daily and secured operations and policy needs; Copernicus further benefits from synergies (*e.g.* on Sentinel 4 and 5) with European meteorological and climate satellites (*e.g.* MTG, EPS-SG) which are planned for the next 20+ years;
- *Weakness*: the EO supply depends partially on non-EU assets (*e.g.*, access to launchers or raw material). Slow products uptake and limited agility compared to competitors, from industry in particular. Lack of integration between components (*e.g.* computing, data, services, in-situ networks), also prevents the full and timely exploitation and downstream user uptake of EU's investment in Sentinels which is still to be achieved;
- *Opportunities*: the gradual integration of Copernicus and other European EO and big data (*e.g.* IoT, data ecosystems), EU computing assets (*i.e.*, EuroHPC) and science excellence in adding value to EO (processing, analysis, modelling, mining, fusion, AI/ML) are instrumental to the Green Deal and can transform the Copernicus next generation product offer and user experience by leveraging the latest space and digital technologies;
- *Threats*: large private corporations, in particular non-EU ones, are becoming aggressively active on the EO and Copernicus service portfolio landscape, with more fit-for-purpose offers, posing a threat to Copernicus at risk of losing its leadership and user base, and for products to become obsolete. This could even undermine the uptake at policy level. Resilience of EU supply chains is another serious threat to Copernicus operations and evolution.

Some of the above issues are being addressed at higher level (*e.g.*, Critical Raw Material act, Access to Space), others can be the purpose of R&I efforts under the present strategy. To be most effective, a long-term R&I commitment on the Copernicus evolution is required to enable this transformation. The recent developments in the AI and machine learning landscape can influence the SWOT analysis by intensifying threats and unveiling new opportunities. The integration of dedicated AI and machine learning-driven activities would expedite the positive impacts.

Europe has a leading advantage in the area of weather, climate and ocean modelling, thanks to its state-of-the-art observing and modelling capacities. Several Copernicus services directly contribute to and benefits from these capacities (C3S, CAMS, CEMS, CMEMS). The AI/ML is both a threat and opportunity for Europe's leading position in this area, and in related downstream sectors.

⁽⁷⁾ Copernicus Space Architecture Evolution, Expert Group Report, DG DEFIS Unit C3, Nov 2022

2.5 Copernicus evolution

The policy and operational nature of the programme makes it de facto a mandatory focus for strong R&I. EU policies are evolving on a continuous basis, with growing expectation on the programme to deliver on policy uptake as flagged by the European Court of Auditors' report 2021 which, whilst praising Copernicus for *"Technologically, the EU has succeeded in becoming a global player in terms of space-based earth observation..."*, also urged for the *"uptake of its space services to fully capitalise on the significant public investment made"*⁽⁸⁾.

This was further confirmed by the Deep Dive meetings with Commissioner Breton and his political priorities for the Copernicus and New Space to evolve as a digital infrastructure⁽⁹⁾ supporting the Green Deal and the resilience of the EU so that the programme continues providing the best in class environmental, security and emergency services. It is also worth recalling that the evolution of Copernicus is also an explicit requirement in Horizon Europe which: *"shall ensure that the R&I needs of the Union Space Programme ... are identified and established as part of the [HE] Programme's strategic planning"*.

Copernicus Research and Innovation aims at ensuring the highest quality of the services meeting user requirements. The commensurate research and development is typically organized in three tiers:

- Tier 1: R&D on timescales of a few months conducted in-house within the services addressing work on the operational system and its configuration funded by Copernicus;
- Tier 2: R&D on timescales of up to 1-2 years to prepare and test evolutions of the services. This work, funded by Copernicus, is largely outsourced to maximize uptake of mature research;
- Tier 3: medium to long-term R&D on several years focusing on upgrades of the services to address EU and global policy priorities. This work is typically funded by the Framework Programme (currently Horizon Europe), including but not only the 'Copernicus Service Evolution Calls', or national research.

The present document is primarily concerned with Tier 3 research.

Whilst Copernicus is a world-class Earth observation programme, it needs continuous evolution to remain so to address nowadays constantly evolving environmental, emergency and security challenges by leveraging the latest science and technology. To support the Green Deal and Digital Transformation, the evolution of Copernicus is planned along four key priorities:

1. **Agile, resilient, expanded and secure Earth observations streams** ensuring robust and timely Copernicus data provision with enhanced performance and reduced dependencies, building on optimized and cost-effective hybrid constellation (new Sentinels – Sentinel Expansion and Sentinel Next Generation missions resp., Contributing Missions and New Space), taking advantage of the latest development in passive and active Earth observation sensing, autonomous observing systems (UAV, AUV, drones, HAPS, FRM, etc.), ground segments, and navigation and communication technologies towards full digital observation of the Earth at scale.

⁽⁸⁾ [EU space programmes Galileo and Copernicus \(europa.eu\), European Court of Auditors, Special Report No 07/2021](#)

⁽⁹⁾ [Regulation \(EU\) 2021/694 of the European Parliament and of the Council of 29 April 2021 establishing the Digital Europe Programme and repealing Decision \(EU\) 2015/2240 \(Text with EEA relevance\)](#)

2. **Boosting data spaces capabilities, interoperability and AI big data analytics**, enhancing data standards and protocols, dissemination, abstraction, compression capabilities and ontologies, the development and certification of AI/ML-ready data, integrated HPC/Cloud/Edge and data management with Internet of Things (IoT) towards added-value products building on the wealth of information available across all space, ground, and service components enabling the full Copernicus data digital transformation.
3. **Service digital uplifting from data to knowledge and decision making**, integrating the latest observations, Earth system process and component modelling, uncertainty quantification and tracing, interactive digital twins, AI/ML (foundation models, transfer learning, graph-based and generative algorithms) and cloud technologies, improving growing security and resilience needs, supporting EU and Member State policies (*e.g.* biodiversity, climate mitigation and adaptation, food security, environmental compliance and security, urban planning, early warnings, human migrations, zero pollution, cultural heritage, Arctic), as well as the global sustainable agenda.
4. **Effective policy uptake and downstream mass adoption** enabled by Copernicus on-demand approaches transforming the user digital interactive experience, by developing innovative push and pull tools anticipating user needs to transform the Earth observation data portfolio ecosystems into a dynamic space service offer, encouraging opportunities across EU space assets (*e.g.* Galileo/EGNOS/GOVSATCOM/IRIS2), promoting Copernicus in all relevant policy and societal areas across sectors in Europe and internationally, driving economic growth and contributing to the wellbeing and security in Europe.

3 Service priorities

3.1 Copernicus Climate Change Service (C3S)

Vision and mission

The mission of C3S is to support adaptation and mitigation policies of the European Union by providing consistent and authoritative information about past, present and future climate. The societal demand for operationally produced insights about the climate is on a steep rise. The service provides free and open access to climate data and tools based on the best available science. Responding to the evolving requirements of our core users, C3S helps them meet their goals in dealing with the impacts of climate change.

Main portfolio and expected evolution

The portfolio of C3S dwells on observed variables chosen according to the concept of Essential Climate Variable (ECVs) as defined by the Global Climate Observing System (GCOS). These variables are used to monitor how the climate has been changing over the last decades but also how it will change in the future. Aside the products directly based on satellite observations (including an increased role 'in a climate sense' of Copernicus Sentinel data), C3S also offers access to global and regional climate reanalyses which integrate the satellite data with our physical understanding of the climate system. These reanalysis products are a key entry in a huge variety of climate adaptation projects and activities. The portfolio is enriched by the access to the multi-system seasonal predictions, the CMIP climate projections and the regional projections of CORDEX, in particular for Europe. All these products and datasets are supplied through the Climate Data Store (CDS) in a way that facilitates the combination of datasets and the generation of new products. The evolution of the portfolio will include the development of a much-enhanced global reanalysis (ERA6) and possibly regional reanalyses, an enriched set of climate predictions (at decadal, seasonal and sub-seasonal timescales), and the operational update of the IPCC climate atlas including strategies for the operational provision of the underlying data, its quality assessment, and recommendations for its use, application and limitation. The evolution of the service offering will also include a near real time climate attribution service focusing on heat stress.

Target policies, customers, users

C3S serves a substantial number of policy domains encompassed by the European Green Deal. These include climate mitigation and adaptation (UNFCC Paris agreement, DG CLIMA, DG ENER, Mission on Adaptation, DG ENV, EEA, JRC, EUCRA), health (DG SANTE, Climate and Health Observatory), renewable energy (DG ENER, DG CLIMA, ENTSO-E, etc.). C3S is also establishing a National Collaboration Programme modelled on the example of CAMS to allow a good articulation and flow of information between the European and national levels for public institutions, fostering collaboration and co-design. The Climate Data Store (CDS) in June 2023 reached 200.000 registered users to whom it typically provides some ~150 TB of quality-controlled climate data per day. Major developments are in progress to upgrade the CDS in the form of a new Climate and Atmosphere Data Store (CADS) system, which will be common with CAMS and will offer increased cloud/edge computing capacity and a much more user-friendly, flexible, and powerful set of tools for developing applications.

Strengths, weaknesses, opportunities and threats

Strengths: C3S provides access to a vast array of quality controlled and operationally produced climate data, that has no equivalent in the world. This extensive data collection allows for robust analysis and assessment of climate trends and changes, and it is one of the reasons why the service and its reanalyses ERA5 featured so prominently in the latest assessment report of the IPCC. The service delivers timely and up-to-date information on climate variables, ensuring that users have access to the most recent climate data available. It also offers long-term datasets, enabling the examination of climate patterns and changes over extended periods.

Weaknesses: The field of climate science and data analysis is rapidly evolving, driven by technological advancements. Climate change is inherently a complex and politically charged issue. C3S operates within a policy landscape that may undergo changes due to shifting priorities (*e.g.*, claims related to extreme event attribution) or geopolitical factors (energy independence in a changing climate).

Opportunities: the effectiveness of the programme relies on its ability to engage and communicate with diverse stakeholders effectively, something that consequently needs to be supported, maintained and whenever appropriate and possible expanded. Leveraging new technologies, adapting to new policy frameworks, navigating international collaborations, and addressing well potential political pressures will be key for C3S.

Threats: failure to adapt and incorporate new technologies could hamper C3S' ability to deliver state-of-the-art services in NRT. C3S will need to stay at the forefront of emerging technologies (*e.g.*, ML/AI) and methodologies to effectively process and analyse vast amounts of climate data.

Short-term (2025-2027) R&D priorities

The most pressing R&D activities needs are in the following domains:

- The clear European leadership on numerical weather prediction made the European global reanalyses the reference point for climate monitoring and -at the same time- the backbone for machine learned weather forecast systems. It is essential for Europe to sustain the global lead in reanalysis and exploit Sentinel data via sufficient resourcing, given the current pace of technology developments. Machine learning can be used in various ways to improve data assimilation as well as post-processed products combining model and satellite data.
- Pure machine learned models show astonishing capabilities in predictions, but also weaknesses when keeping the model state uniform across forecast lead-time (instead of blurring with time) and when extrapolating into unknown climate states. Hybrid approaches – that combine conventional models with machine learned tools – may play an important role in providing timely information about the evolution of the climate on a seasonal to decadal timescale.
- The growing impact of climate on our society calls for novel ways of linking climate drivers with climate response, adaptation options and mitigation policies. Approaches such as story lines that start from user decisions, as those developed within the World Climate Research Programme (WCRP) 'My Climate Risk' lighthouse activity and a-posteriori attribution should be enhanced and complement forecast-based activities.

Long-term (next MFF) R&D priorities

Topics for longer-term research include:

- Continued investment of application of AI/ML to climate prediction and projections, post-processing, data assimilation and forecasting systems.
- Developments to progressively integrate novel products from Copernicus Sentinel data, which are now gradually generating long term climate data records expanding the offering with respect to ECVs.
- Development of the Earth System Models (e.g. IFS, ICON) to better represent the processes in the climate system & uptake of Digital Twin developments for C3S application.
- Exploration of novel ways of linking societal impacts to the modulation of climate impact drivers on an interannual and multi-decadal timescales, and associated storylines and communication.

Key R&D partnerships at EU and International level

C3S largely builds on the whole European R&I community working on climate observations and predictions. Besides C3S development contracts and relevant Horizon projects, most other collaborations are worldwide. It is not possible to be exhaustive, but we can mention the following: the WMO Global Framework for Climate Services (with a key role for C3S to act as an example of an operational climate service), the Global Climate Observing System (for which C3S act as an official reference repository), the IPCC (where ERA5 is mentioned over 240 times), WCRP, EUMETSAT for the underpinning work to provide the backbone of the reanalysis products, ESA for the underlying work under its Climate Space (former Climate Change Initiative) programme, ...

3.2 Copernicus Atmosphere Monitoring Service (CAMS)

Vision and mission

CAMS provides consistent and quality-controlled information related to air pollution and health, energy, greenhouse gases and climate forcing, everywhere in the world. This is done by means of value-adding information workflows combining data from satellite and in-situ observations and numerical models, to serve users in the public and commercial domains. CAMS also operates the new Greenhouse Gas Emissions Monitoring and Verification Support Capacity, which will provide pivotal input to the EU Green Deal and Paris Agreement process.

Main portfolio and expected evolution

CAMS provides open & free information products based on Earth Observation about past, current and near-future (forecasts) global atmospheric composition; the ozone layer; European air quality including tools co-designed with users in the policy domain; emissions and surface fluxes of key pollutants and greenhouse gases; solar radiation; and climate radiative forcing. Most of the products are existing and available from the atmosphere data store (ADS), but they will be strengthened with use of upcoming Copernicus satellite missions, particularly Sentinel-4, Sentinel-5 and CO2M. Major developments are in progress to upgrade the ADS in the form of a new system (CADS), which will be common with C3S and will offer increased cloud computing capacity.

Target policies, customers, users

CAMS serves a substantial number of policy domains encompassed by the European Green Deal. These include air pollution (DG ENV, EEA, JRC, Air Quality Directive, National Emissions Ceiling Directive, etc.),

health (DG SANTE, DG HERA, Climate and Health Observatory), renewable energy (DG ENER, DG CLIMA, etc.), ozone layer (DG CLIMA, WMO-UNEP Montreal Protocol, etc.), greenhouse gas emissions (methane strategy, UNFCCC Paris agreement, DG CLIMA, DG ENER, etc.). CAMS has established a National Collaboration Programme to allow a good articulation and flow of information between the European and national levels for public institutions, fostering collaboration and co-design. Finally, CAMS has a very active and growing user community (about 30k users, 3k routinely active) reaching in the end over 200M people daily through broadcast TV, smartphone applications and websites. CAMS direct user typology is broadly: research (60%), commercial (20%), public (10%), not known (10%).

Strengths, weaknesses, opportunities and threats

Strengths: CAMS has very few equivalents in the world for most of its workflow/systems/application areas. A key strength has been continuous effort to advance the systems beyond the state-of-the-art and to rapidly harness the benefits from new EO capabilities.

Weaknesses: the needs from users, particularly in the policy context (largely encompassed within the EU Green Deal but not only), are pressing so targeted R&D is essential to expedite availability of new/improved products. Novel instruments may present some challenges in using them optimally in CAMS systems.

Opportunities: CAMS is fortunate to have three upcoming Sentinels to underpin its activities: Sentinel-4, Sentinel-5 and CO2M (and strong interest for some Copernicus Contributing Missions). Another aspect is the development of AI/ML, which will likely drive substantial changes in the information workflows. There also, it is of importance that the science community relevant for CAMS can “change gear” through targeted projects.

Threats: there are some risks associated with delays in launch dates and availability of fully qualified level-2 data for use in CAMS operational workflows.

Short-term (2025-2027) R&D priorities

The most pressing R&D activities needs are in the following domains:

- Research activities to develop new integrated soil-vegetation-atmosphere modelling and data assimilation of deposition fluxes and of volatile organic compounds and pollen emissions, exploiting latest developments on CLMS where appropriate;
- Research into the impact of wildfires on vegetation. The overall objective is to enable accounting for the interaction between droughts, fires and vegetation in the CO2MVS system as well as to improve the estimation of fire emissions in CAMS, the forecast of their evolution (next days) and the fire risk forecasting in CEMS;
- Advanced use of AI/ML for atmospheric composition applications including correction of state dependent model errors; enhancing parameter tuning/estimation and improving data assimilation in general; explore ensemble forecasting based on machine learned atmospheric composition models;
- Advanced activities to quantify the impact of indirect effects of aerosols and their interactions with clouds. These processes are not well represented in weather and climate models, and a better understanding will be pivotal (including in an urban context) to be able to mitigate human induced changes and improve weather (including extreme) forecasting;

- Improvement of aerosol data assimilation with a focus on better speciation and size distribution specification by using new generation of aerosol satellite sensors, such as 3MI and MAP onboard CO2M;
- To support long period (20th century) reanalysis in an Earth System framework, focusing for CAMS on aerosol, tropospheric chemistry and stratospheric chemistry. Explore use of so far unused satellite data (especially in the VIS regarding aerosol, which have been available since the 1980s);
- Research on developing/improving information products regarding greenhouse gas emissions from the AFOLU (Agriculture Forestry and Other Land Use) sector from combining atmospheric observation, land-use observations and Digital Global Vegetation Models (DGVM);
- Expand uncertainty estimates on the CAMS portfolio and associated Evaluation and Quality Control;
- Research on exploiting EO-based (existing and planned) observations of GHG (*e.g.* CO₂, CH₄) and other atmospheric pollutants in CAMS inversion systems to improve anthropogenic and natural emission quantifications, to detect hotspots and anomalies, and to support environmental and climate policies from urban to global scale.

Long-term (next MFF) R&D priorities

Topics for longer-term research include:

- Continued investment of application of AI/ML to atmospheric composition data assimilation and forecasting systems and the dissemination of these services to the public;
- Development of the IFS & uptake of Digital Twin developments for CAMS applications;
- Improvement of Digital Global Vegetation Models, especially in the context of observation-driven operational services;
- Improvement of plume inversion estimation algorithms (for emissions estimation) based on new generation of satellite data.

Key R&D partnerships at EU and International level

CAMS largely builds on the whole European R&I community working on atmospheric composition at large. Besides CAMS development contracts and relevant Horizon projects, most other collaborations are worldwide. It is not possible to be exhaustive, but we can mention the following: WMO's Global Greenhouse Gas Watch (with a leading role of CAMS/Copernicus regarding modelling and data assimilation aspects of the initiative); International Methane Observatory; CEOS-CGMS activities on Greenhouse Gases; Wildfire monitoring activities with several country including in the context of IGAC; collaboration on air quality forecasting with the US, Canada, Korea and many other countries.

3.3 Copernicus Emergency Management Service (CEMS)

Vision and mission

CEMS provides information in support of and in coordination with public authorities concerned with civil protection, supporting civil protection and emergency response operations (improving early warning activities and crisis response capacities), and prevention and preparedness actions (risk and recovery analyses) in relation to different types of disasters as well as national and regional environmental agencies and services. It covers with its products and data all phases of the disaster risk management cycle, *i.e.*,

prevention, preparedness, response and recovery. CEMS aims to increase the efficiency of emergency services and to maintain the EU's autonomy and leadership in disaster risk management.

Main portfolio and expected evolution

CEMS currently consist of the following components: (1) *On-demand mapping* component which provides geo-spatial information in support of emergency management (rapid mapping mode) or in support of prevention and recovery (risk and recovery mapping mode); (2) *Early warning and monitoring* component for floods (EFAS, GloFAS, GFM), forest fires (EFFIS) and droughts (EDO, GDO); (3) *Exposure mapping* component which provides highly accurate information derived from satellite and in-situ data on the presence of settlements, its related infrastructure and population based on GHSL. In the next few years CEMS will continue to evolve these components through continuously adding new data sources (*e.g.*, new Sentinels with higher resolution), new data from the Copernicus Contributing Missions, new satellite sensors data from space agencies such as NASA, CSA, JAXA, *etc.*, more in-situ data), improving models and methods (*e.g.*, integration of AI based methods, increase of model spatio-temporal resolutions, data fusion, *etc.*) and adding, based on user requirements, new and innovative products (*e.g.*, expanding forest fire products to the global scale, adding coastal flood risk products, enhanced drought and detection, prediction and attribution, *etc.*).

Target policies, customers, users

CEMS key users are mainly national authorities related to disaster risk management, which includes civil protection authorities, hydro-meteorological services and environmental authorities amongst others. At international level DG ECHO's Emergency Response Coordination Center (ERCC) but also other actors such as EEAS, UN agencies, international development and aid organizations (*e.g.*, Red Cross) are the main users. The Union Civil Protection Mechanism, Floods Directive, EU Solidarity Fund, the Green Deal, Forestry related initiatives (*e.g.*, EU Forest Strategy, EU Biodiversity strategy, EU forest monitoring framework, *etc.*) and the EU international aid policies form the main European policy framework that CEMS is supporting. At global level the Sendai Framework for Disaster Risk Reduction, the Paris Agreement, the Sustainable Development Goals and the New Urban Agenda are the most important policy frameworks for CEMS.

Strengths, weaknesses, opportunities and threats

Strengths: CEMS has established a very close relationship with its key users through regular meetings and workshops, expert groups, trainings and user surveys over the past years. The current set of CEMS products has been developed taking these detailed user requirements into account and has led to a high acceptance and recognition of CEMS.

Weaknesses: Currently, the uptake of new research into operational CEMS products is limited as most of the relevant research is not mature enough or does not have the necessary technology readiness level that is required for an operational emergency management service, including its 24/7 components.

Opportunities: New capabilities of future Sentinel and contributing missions, new open data, new cloud and processing infrastructures (CDSE, EuroHPC, EOSC), Artificial Intelligence and new production methods represent a great opportunity to further evolve CEMS. More awareness regarding products can attract new R&D and promote new services.

Threats: As CEMS has seen a constant increase over the past years of On-Demand activations there is a risk, that if this trend continues and no sufficient budget is available, priorities for the activation requests may have to be implemented.

Short-term (2025-2027) R&D priorities

- Automated characterisation of building height and building use (*e.g.*, residential, industrial, commercial, public, poor/rich) through integration of different sensor types (*e.g.*, optical, radar, night-time lights) and IoT;
- Integration of new sensors (incl. of 3D data derived from multi sensor platforms) for active delineation, grading and damage assessment for global fire detection and fire monitoring from polar-orbiting and geostationary sensors, including refined post-processing for active-fire detection confidence and false alarm removals;
- Integration of high and very-high spatial resolution data and sensors for continuous multi-scale mapping and assessment of fuel structure and condition at pan-European level, including active (SAR, Lidar) and passive (multispectral and hyperspectral) remote sensing data;
- Improvements of the hydrological predictions for the flood (including flash floods) and drought early warning and monitoring component through data assimilation and/or multi-objective parameter calibration and regionalization using satellite based or in-situ data;
- Improvements of the hydrologic process representation in the continental or global scale hydrologic model of the flood and drought early warning and monitoring components;
- Methods for addressing limitations of Synthetic Aperture radar (SAR) based flood monitoring (Urban areas, dense vegetation, ...);
- Enhanced seamless sub-seasonal to seasonal predictions of severe-to-extreme drought events and associated multi-sectoral impacts with a focus on the water-food-energy nexus;
- Optimised integration of different data sources (*e.g.*, reanalysis + observations from rain gauges + remote sensing) and different indexes characterising droughts;
- Integration of UAV along the full value-added chain (*i.e.*, data planning, flight, data acquisitions and processing) in the current emergency response operations for improving the thematic accuracy of the damage assessment;
- Advanced drought event tracking methods;
- Innovative approaches for multi-sectoral risk and impact assessments related to drought, heatwaves and compound (drought/heatwaves) events;
- Integration of machine learning methods in hydrological modelling and remote sensing to improve the monitoring and forecasting of floods and droughts.

Long-term (next MFF) R&D priorities

- Integration of high and very-high spatial resolution data and sensors for continuous multi-scale mapping and assessment;
- Integration of satellite gravimetry information to better represent continental and large-catchment water storage regarding flood and drought risk;
- Performing dedicated studies on alternative methods, data, workflows for all CEMS components to be included in operations;
- Developing and testing new products (*e.g.*, Global Wildfire Information System, provision of day/night population grids), and prototyping of innovative tools for product dissemination;
- Development of high-resolution dynamic heatwaves detection and monitoring with a focus on urban areas;
- Optimally exploit and integrate data from the Copernicus Expansion and Sentinel Next Generation Missions in the existing CEMS portfolio;

- Continued enhancement, further development and quality assessment of the models used in CEMS including increase in spatio-temporal resolution, process representation and integration of machine learning approaches;
- Uptake of Digital Twin developments for CEMS.

Key R&D partnerships at EU and International level

Continue and enhance the synergies with other Copernicus services. Ensure strong linkage to Horizon Europe and continue cooperation activities with European research institutions and organisations. Continue embedding CEMS in international initiatives such as the International Charter Space and Major Disasters, the Group on Earth Observations and close linkages to space agencies such as ESA, NASA, *etc.* and other relevant key partners such as WMO and relevant UN agencies.

3.4 Copernicus Land Monitoring Service (CLMS)

Vision and mission

CLMS aims at reaching a truly dynamic monitoring of our changing terrestrial environment by providing users with timely, on-demand, analysis-ready and advanced products on EU and global eco- and anthropogenic systems (urban, agriculture, forest, lake, river, wetland, ice, snow, managed and non-managed land, *etc.*), as well as quantity and quality of their natural resources, such as biomass, biodiversity, water quality, soils and minerals, through bio-geophysical parameters and derivatives of them.

Main portfolio and expected evolution

The CLMS portfolio is comprised of the following main product groups: Land cover and land use monitoring and mapping; Thematic hot spot mapping; Biophysical parameters; Reference data and image mosaics; and the European Ground Motion Service. Evolutions are focused on improvements to the algorithm in terms of processing (timeliness, scale, coverage), higher frequency and more detailed thematic information, input, accuracy and uncertainty traceability.

Target policies, customers, users

The CLMS addresses the data and information needs of a wide range of policies such as environment including biodiversity, climate change, agriculture and forestry, regional and urban planning, as well as water management, transport, energy, or neighbourhood policy, at global (UN organisations, NEAR, INTPA, ECHO, ENV, AGRI), European (DEFIS, ENV, CLIMA, REGIO, AGRI, MOVE, ENER, ...), national, and local level. The European Green Deal addresses nine policy objectives where the needs for fit-for-purpose information on land condition, land cover and land use change define the strategic direction for the CLMS.

Strengths, weaknesses, opportunities, and threats

Strengths: CLMS has accumulated extensive and valuable knowledge concerning the production of detailed high-quality land information based on space and in-situ data meeting key user requirements. Products are fully documented, quality controlled, validated and reviewed.

Weaknesses: The production of key high-quality CLMS products requires a substantial amount of manual intervention leading to high production costs, complicated production chains, and long response

time. Uncertainty traceability is often missing for assessing where products can be improved. Spatial resolution of some products lacks relevance for small EU Member States.

Opportunities: New production methods based on long timeseries combined with *e.g.*, AI techniques, next generation Sentinels, upcoming expansion missions, and new contributing missions in combination with Europe's open data strategy and new cloud and processing infrastructure (CDSE) would enable the CLMS to meet known and emerging user requirements related to product update frequency and thematic content. Enhance collaboration between Copernicus services and with R&D communities; serving local user needs, create a 'rapid' research mechanism with a (very) short demand-action cycle.

Threats: Administrative steps necessary for implementation can have an impact in operationality if delays occur, impact of inflation in the costs of a fully externalised production, increased costs related to high volume of data being processed and stored, increased complexity to access the data affect the user uptake of products. Research cycles don't follow service (contracting) cycles, creating delays or impossibility to operationally implement results. Rapid research response to the needs and flexibility (agile) approaches are not commonly available.

Short-term (2025-2027) R&D priorities

The CLMS short-term R&D priorities focus on:

- finding solutions to the well-known user demand for more timely products with higher spatial and temporal resolution and lower latency;
- development and validation of new and improved production methods *e.g.*, as regards automation, the use of AI techniques, efficient workflows, and optimal use and combination of available data from Sentinels and Contributing Missions and in-situ data to also address needs of all Member States, including the smaller ones;
- prepare the CLMS product and service portfolio for the upcoming Copernicus expansion missions to be able to fully exploit new data with higher spatial resolution, additional spectral and thermal information, L-band SAR, and data fusion;
- investigate the potential to consolidate and enlarge the CLMS portfolio by proposing Arctic mapping products, focusing on reliable and consistent land use information, and addressing thematic gaps;
- new retrieval algorithm for non-unique "water" types; Global wetland mapping algorithms; Emissivity 'land' database for Land Surface Temperature with LEO sensor; Ensemble products (*e.g.*, for vegetation monitoring); Forecasting capacity/approaches;
- on the user uptake aspect, develop a CLMS National Collaboration Programme to foster the uptake at national level, receive feedback on the products and ensure adequate alignment of the portfolio to the user needs;
- conduct research in support of the development of a Copernicus Land Data Store (CLDS).

Long-term (next MFF) R&D priorities

- Optimal exploitation and integration of data from the Copernicus Expansion and Sentinel Next Generation Missions in the existing CLMS portfolio;
- Modelling at various timescale (back and forward), as well as integrated sectoral products (*e.g.*, for water management, biodiversity, land use trade-offs, *etc.*). CLMS until now focused more on observation but it is expected in the future to start moving towards modelling and prediction;
- Further research regarding known and emerging thematic gaps in areas such as terrestrial biodiversity, wetland information, urban applications, vegetation height, carbon uptake/biomass; soil composition and moisture, and water balance information.

Key R&D partnerships at EU and International level

JRC and the EEA work closely together and with other Copernicus Entrusted Entities. The EEA relies to a very high degree on the European Environment Information and Observation Network (Eionet) and eight centres of thematic expertise contracted by the EEA, called European Topic Centres (ETCs). Future partnerships can also be established with the European Research Infrastructures. Engagement with CEOS LSI-VC and EU-US Land Imaging collaboration are also key.

3.5 Copernicus Marine Environment Monitoring Service (CMEMS)

Vision and mission

The Copernicus Marine Service provides the EU with a world-leading, reference digital information service on the world ocean and EU regional seas in their physical, biogeochemical and sea-ice dimensions, from the surface to the deep ocean. Copernicus Marine values the EU excellence in marine sciences, technologies and services, and aims at fostering evidence-based decisions for a sustainable ocean. Copernicus Marine capacity for authoritative information and services on the marine environment extends from ocean climate to near-real time monitoring and forecasts. With the ocean being higher and higher on political agendas and the EU occupying a growing place on the international scene, CMEMS directly contributes to EU's marine and maritime-related policies. Copernicus Marine also supports applications dealing with *e.g.*, maritime safety, sustainable use of marine resources, healthy waters, informing coastal and marine hazard services, ocean climate services, protecting marine biodiversity.

Main portfolio and expected evolution

The Copernicus Marine Service monitors in real time and over the past decades the world ocean across the entire water column using in-situ and satellite observations and monitoring and forecasting systems (*i.e.*, numerical models with data assimilation). It provides regular and systematic reference information on the physical (blue) and biogeochemical (green) ocean and sea-ice (white) state for the global ocean and the European regional seas.

Copernicus Marine will progressively embrace the new capabilities of digital services in synergy with the Digital Twin Ocean development carried out as part of the Mission Ocean and Destination Earth initiatives. Copernicus Marine is also preparing the evolution of its capacity and portfolio through the implementation of future EO missions, the development of the next generation of ocean monitoring and forecasting systems (improved Arctic monitoring, ensemble and extended-range forecasting, higher resolution, longer reanalyses) and the preparation of new services for the coastal marine environment (with Member States), for marine biology and biogeochemistry and for climate projections dedicated to the marine environment.

Copernicus Marine provides reference ocean and coastal marine information for the other Copernicus services (C3S, CAMS, CEMS, CLMS).

Target policies, customers, users

The Copernicus Marine answers to the increasing needs of improved 3D ocean monitoring and prediction capabilities to understand, predict and adapt to climate change, for a sustainable management of the ocean and its resources, a sustainable blue economy and to better protect marine ecosystems which includes its links to biodiversity and biogeochemistry. The Copernicus Marine Service is a powerful instrument to support the European Union in the Green Deal, Digital Strategy and the environmental policies implementation (Marine Strategy Framework Directive, EU Arctic Policy, Maritime Spatial

Planning, Common Fishery Policy, Water Framework Directive, Habitat Directive, EU Biodiversity Strategy), while playing a pivotal role in international initiatives (e.g., UN Decade of Ocean Science, UNESCO-IOC). The Copernicus Marine has established a National Collaboration Programme with European Member States to allow i) the co-design of a seamless offshore to coastal ocean monitoring, through User Engagement Call for Tenders and ii) a good articulation and flow of information between the European and national levels to foster collaboration and co-design, through its Marine Forum. Finally, The Copernicus Marine has a very active and growing user community (about 55k users, 3k routinely active) reaching over 700k people yearly through the Copernicus Marine website.

Strengths, weaknesses, opportunities, and threats

Strengths: the Copernicus Marine Service has set global standards for ‘research-to-operations’ and ‘operations-to-research’ processes. The service is world-leading through its coverage and comprehensiveness, its balance between state-of-the-art science and operational commitments, and the consistency of its portfolio where satellite observations, in-situ observations, and 3D model simulations are coherently proposed to describe the ocean. This requires steady collaboration and innovation across research and technology, in observations, modelling, assimilation, reporting and product and service delivery.

Weaknesses: the service strongly depends on in-situ observations to observe the ocean interior. CMEMS must evolve to remain state-of-the-art and answer to the increased ocean monitoring and prediction needs. It must inform and embrace the new capabilities of future Sentinel missions, digital services (HPC, Cloud) and Artificial Intelligence.

Opportunities: CMEMS is well positioned to develop user-friendly data catalogue and visualisation tools, use cases, capacity building and learning activities to facilitate the last mile data uptake for policy applications by DGs, Member States and Regional Conventions; to promote new cloud-based Marine Data Store, offering users expanded functionalities to analyse and post-process data directly on the cloud and browse data across all temporal and spatial directions; to gradually build capacities in Europe and in Africa as a key priority outside Europe.

Threats: The sustainability and evolution of the global/regional and coastal in-situ networks is a major issue and lack of adequate actions is a threat for the Copernicus marine service evolution. MOi works with EuroGOOS and EEA in the framework of the European Ocean Observing System (EOOS) to document needs and advocate for the evolution of in-situ observing capabilities.

Short-term (2025-2027) R&D priorities

For the period 2025-2027, the main priority topics for the Tier-3 Horizon Europe R&D activities are:

- Improving ocean reanalyses, analyses and forecasts through evolutions in marine data assimilation techniques: (i) coupled data assimilation issues (e.g., between ocean and biogeochemistry, ocean and sea-ice, ocean and waves and atmosphere, coastal ocean and estuarine/inland waters), (ii) development of multi-scale methods, (iii) ensemble methods, probabilistic forecasting and extended range forecasts (monthly) and (iv) the use of new observations (e.g., future satellite missions, and new in-situ observations);
- Improved models for the Arctic Ocean with a better representation of the interactions between ocean, waves, sea-ice, biogeochemistry and marine ecosystems. Development of sea-ice models based on a more realistic rheology, with improved representation of sea-ice thermodynamic and dynamic processes. Improved methods to assimilate future sea-ice observations (future Sentinel missions: CIMR, CRISTAL);
- Use of artificial intelligence methods to generate short-term (e.g., 10-day) ocean and sea-ice forecasts. Validation of forecast skills and comparison of performance with state-of-the-art ocean

and sea ice forecasting models; Development of hybrid approaches (combining conventional numerical models with AI tools) for ocean and sea-ice models.

- Improved downscaling capacity in ocean forecast models including physics, biogeochemistry, sea ice and data assimilation using advanced methodology based on two-way nesting and/or machine learning approaches;
- Improved carbon modelling/data assimilation at regional and global scales for the provision of multi-year products to better address carbon-cycle applications, including carbon accounting;
- Development of capabilities for producing seasonal forecasts of marine environment changes from global to regional scales (physics, biogeochemistry, biology, high trophic levels incl. main exploited fishes);
- Development of capabilities to predict and detect extreme events of the marine environment over the past decades and in near-real time (*e.g.*, marine heat waves, rogue waves, eutrophication, harmful algae blooms, extreme coastal sea levels, *etc.*) and to attribute them to climate change and other human activities, including to support reporting activities.

Long-term (next MFF) R&D priorities

The Copernicus Marine in the post 2028 period will fully benefit from the Digital Twin Ocean infrastructure. The above R&D needs will still apply but we anticipate more efforts on Artificial Intelligence (all along the value chain from raw data to information), hybrid approaches combining numerical models and AI and use of new HPC infrastructure. The uptake of the Sentinel expansion missions for the Arctic (CIMR, CRISTAL, ROSE-L), CO₂ monitoring (CO₂M) and the coastal ocean (CHIME, LSTM) will require specific R&D needs. The same holds for the preparation of the uptake of Sentinel NG missions. Similarly, Copernicus Marine will exploit the main evolutions of the in-situ observing system (*e.g.*, One Argo, Marine Life 2030, marine biodiversity observations).

R&D support to our national collaboration programme for coastal zone monitoring and forecasting (seamless monitoring from the open ocean down to the coastal zone and estuaries fully taking into account Digital Twin Ocean infrastructure and capabilities) should also be considered. New user requirements and R&D topics for the evolution of the Arctic and Coastal Copernicus Thematic Hubs are likely to emerge in the coming years. These could be implemented through dedicated R&D activities in CMEMS.

Key R&D partnerships at EU and International level

In complement to Horizon Europe and Mission Ocean, R&D activities led by ESA, EUMETSAT, JPI Oceans & Climate and Sustainable Blue Economy Partnership (SBEP) are of importance for the Copernicus Marine. The United Nations (UN) Decade of Ocean Science is a major undertaking at international level to advance the research and innovation for the ocean. The UN Decade Collaboration Center for Ocean Prediction led by MOi has direct links with Copernicus Marine.

In addition, co-construction approaches of R&I with stakeholders should be further developed for Copernicus Marine new offers to be better understood, useable and used by marine practitioners, and for R&I to be ultimately more effective in terms of resilience and governance.

3.6 Copernicus Security Service (CSS)

Vision and mission

The Copernicus Security Services (CSS) provide timely information to support the Union's security, safety, law enforcement, as well as EU international policies and commitments through three dedicated Earth Observation-based services: Copernicus Maritime Surveillance (CMS), Copernicus Border Surveillance (CBS), and Support to EU External and Security Actions (SESA). The services are available only to Institutional players in the European Union or to third parties under specific conditions justified along related EU policies.

At its core, the Security Service exploits a network of satellites (Sentinel and Copernicus Contributing Missions) combined with ground-based data and information, and rely on data processing systems to generate added-value information for security applications. Security services are very demanding in terms of observation requirements, usually requiring short acquisition to delivery times during periods that may not coincide with satellite overpassing. They also often require fusion with other information based on local intelligence or in-situ data, to deliver actionable intelligence in response to specific user needs.

Earth Observation (EO) technologies have advanced significantly, both in terms of observation and data processing capacities, as well as commercial delivery models, allowing for narrowing the gap between user expectations and service delivery. As such it became key to incorporate new technologies into operational services, rendering them more fit for the purpose they were designed for, while countering obsolescence.

The EU Space Programme Regulation, which establishes the rules and objectives for the EU space program for the period 2021-2027, came into force in 2021 and marks the beginning of the second phase of CSS operations. It also underlines the importance of ensuring the evolution of CSS and reinforcing the link with Horizon Europe to meet the evolving security needs and challenges of the EU.

Due to the operationally oriented nature of the Agencies under the Copernicus Security agreements, it was necessary to complement the effort done through dedicated R&D activities aligned with service evolution needs, while maximising the probability of uptake of R&D results and foster Industrial competitiveness.

To meet the challenges above, DG DEFIS and DG JRC, in coordination with the Copernicus Committee, User Forum and Entrusted Entities, joined efforts since 2022 to enhance the streamlining of research and innovation based on user demand and operational requirements. This joint effort materialises through a **Strategic Research Agenda** (CSS-SRA), with the following objectives in mind:

- Ensure that the Copernicus Security Services evolve by utilizing state-of-the-art data, technologies and tools to deliver timely information in response to evolving user needs and security challenges.
- Expand the service scope and increase user uptake, by improving fitness for purpose, complement the existent service portfolio, integrate new technologies.
- Foster EU resilience, strategic and economic autonomy, and Industrial competitiveness.

The CSS-SRA is published on a yearly basis setting the medium-term priorities for R&D and detailing actionable actions for the near-future. It also promotes dialogue among stakeholders and the exchanges

of best practices, avoids unnecessary duplications and promotes technology spin-off to operational environments.

Main portfolio and expected evolution

Copernicus Maritime Surveillance (CMS) service

The CMS service provides EO products (satellite images and value-added products) to support monitoring of human activity at sea, covering seven function areas: maritime safety, maritime security, customs, law enforcement, marine pollution monitoring, fisheries control, and international cooperation.

The service offers a wide geographical coverage based on high-quality earth observation data combined with maritime surveillance information. Users can obtain coverage in areas of European interest worldwide, with varying resolutions and sensor types. Additionally, the CMS service forms one component of the broader maritime picture offered by EMSA, allowing users to access other complementary datasets, such as vessel position reports, and selectively combine the data and integrate the data into their decision-making process.

CMS looks further to explore emerging AI and ML technologies and introduction of the New Space Data to increase coverage, detection capacities, response time, and quality of services in general.

Copernicus Border Surveillance (CBS) service

The CBS service offers to its end-users a set of products based on EO data enriched with other data sources. By means of the Copernicus products, Frontex provides support to national border management authorities in managing irregular migration and combatting cross-border crime in line with the European Integrated Border Management.

CBS offers 16 products categorised under three main groups: Reports; Map and Cartographic Products; Digital and Interactive Products.

Currently, Frontex is moving from a general service approach towards a product centric approach. The transition has been introduced as a result of lessons learnt and users' feedbacks and requirements collected during operations, workshops and survey, over the whole implementation period of the programme.

The CBS evolution efforts are focused on further process automation and data fusion to ensure up-to-date picture of the EU external borders with the final aim to increase internal security and contribute to the EU strategic autonomy.

Copernicus Support to EU External and Security Actions (SESA) service

The SESA service aims to support the EU and its Member States by providing geospatial information and products related to security and crisis management. It focuses on monitoring and analysing activities and events of interest in regions outside the European Union.

The service utilises a wide range of Earth Observation data, including satellite imagery, in combination with other sources of information to provide timely and accurate situational awareness to EU decision-makers and operational actors involved in EU External Action. It is an on-demand service that can be activated to react within very short timelines, as needed to respond to such as political or armed conflicts.

On the other hand, the service can carry out monitoring over more extended periods in order to provide a comprehensive overview of how situation is changing.

The SESA portfolio covers broad application areas, such as (i) Security of EU Citizens; (ii) Crisis and Conflict; (iii) Rule of Law; (iv) Humanitarian Aid; (v) Transport Safety and Security; (vi) Stability and Resilience for Development; (vii) Cultural Heritage; (viii) Environmental Compliance; (ix) Climate Security (incl. Arctic and Polar); (x) Health Security and, (xi) International Trade and Economic Diplomacy.

Target policies, customer, users

Copernicus Border Surveillance (CBS) is an integral part of the EUROSUR regulation. It supports activities of national border management authorities in the Member States that are members of the European Border and Coast Guard. This support encompasses EO services aimed at preventing and detecting cross-border crime at external borders, enhancing border control efforts, and addressing irregular migration. Additionally, it contributes to life-saving operations at sea while also supporting endeavours to prevent and detect cross-border criminal activities. CBS also contributes to the EU's Coast Guard Functions in Frontex, EMSA and EFCA.

Copernicus Maritime Surveillance (CMS), implemented through EMSA, serves a wide range of users across over 60 National organisations, the United Nations and the EU Fisheries Control Agency (EFCA), providing services across a plethora of policies related to environmental compliance, law enforcement, security, safety at sea or supporting EU International commitments. Monitoring oil spills outside EU waters, helping National authorities in detecting Illegal and Unregulated Fisheries, supporting National Navies in capacity building in 3rd countries, or counter-piracy and anti-drug and smuggling operations are everyday examples of what is being addressed by CMS. As such, it serves a series of policies under DG MOVE, DG MARE, EEAS, TAXUD, HOME and clearly also Member States National policies.

The Support to EU and Security Action Service (Sept. 2023) SESA replaced the precursor SEA service and is implemented by the EU SatCen. It offers support to a variety of users involved in security and crisis management activities as well as humanitarian efforts. It's worth noting that the list of users may vary depending on the scope of its operations, regional priorities, and the specific needs. SESA supports EU external policies and international commitments and some EU internal policies along its service portfolio.

Strengths, weaknesses, opportunities, and threats

Strengths: Copernicus security services provide valuable information to users over wide areas with difficult access, either due to remote location or difficult access due to e.g. topography or on-going conflicts. They are normally used as a first screening to plan or re-focus subsequent operations. CSS can save precious resources, e.g. over maritime areas inspecting only those areas where suspicious activity has been detected and, therefore, allowing large savings in unnecessary flyovers.

Weaknesses: a major weakness has been the system reactivity and resolution, often falling well below what is needed for many applications. That is mostly due to the lack of persistency in observations, satellite revisiting times and processing capacities on ground often depending on human interpretation.

Opportunities: The NewSpace sector, advances on processing capacity through e.g. machine learning, provide a clear opportunity to improve the service fitness for purpose. It is key to bring these new technologies to a level of maturity where they can be fully exploited by the services, as well as empowering the users in developing additional capabilities, relying on the 'core' services while adding specific know-how and using complementary information. Technology demonstration in operational scenarios and validation, new processing paradigms for increased reactivity as well as data fusion with

non-EO data are therefore key elements for a R&D strategy for the coming years. This goes together with expansion of the current scope of the services to cater also for specific local and regional needs.

Threats: given its sensitive nature, the security services and associated data are provided on a *need-to-know* basis and related information, if not duly protected, may have severe consequences.

Short-term (2025-2027) R&D priorities

Short-term R&D priorities are specified in the SRA-CSS V1.0 (Aug.2023) published in Sept. 2023. They provide direction for the forthcoming EU Horizon Europe Research and Innovation topic specifically dedicated to Copernicus Security Service (HORIZON-CL4-2024-SPACE-01-36), opening in Q4 2023.

The guidelines complement the information in the original call with the outcome of the SRA concertation process 2022-23, aiming at driving R&D activities along user needs, encourage innovation and avoid unwanted duplication, therefore increasing the likelihood that results will be exploited in an operational context. Two actions are foreseen:

Action A: Innovative methods and technologies to explore new and enlarged data sets and data processing paradigms

This action should target the identification and integration of complementary data sources for end-user's benefit, while reducing products delivery times through streamlining data processing and testing new EO paradigms for proactive monitor, including those derived from the commercial sector.

By fostering the integration of new technologies and data in the security service, Europe will be more prepared to respond to the evolving challenges concerning its Strategic and Industrial autonomy.

The work to be performed during the project lifecycle should be steered along medium (2025-2027) to longer term (2027+) objectives, building on the existing know-how and setting the grounds for follow-on activities in the near future. For this specific action, a phased approach is expected, focussing on the following elements, or a subset:

- Mapping of existing and planned relevant technologies and data, with special emphasis on EU assets, with an analysis of the potential measurable impact in operational (realistic) scenarios – and identification of gaps, in particular:
 - NewSpace (new constellations and sensors)
 - Identification of relevant additional datasets along services and additional user needs
- A strategic approach for the exploitation and/or development of algorithms for automated processing of information aiming at
 - Automation of tasks for product generation in core services, aiming at increased reactivity and ability to process large datasets
 - Analysis of AI models applied to EO applications and their adaptation to the specific case of security applications
 - Algorithms (also AI) for the identification of additional sources of public information and datasets relevant to security applications
- Extraction and fusion of additional data sets for added-value services, such as publicly available information, social media or relevant data derived from other Copernicus Security services (*e.g.*, Climate Change, Marine). Demonstration actions and proposed deployment plan.

- Demonstration and validation of technology in operational environments, in particular new space and data processing paradigms. Demonstrations should aim at fast reaction, added-value and EU strategic autonomy, while being realistic in terms of cost/benefit;
- Deploy a concept of *rapid reaction*, including pre-emptive tasking over zones with higher conflicts risk, aiming at near-real time processing and delivery;
- Propose a realistic strategy and exploitation plan for deployment of mature (TRL>6) elements for 2027+.

Proposals should be fully complementary to what the current Security services already provide (*Section: Introduction to the Copernicus Security Service*), with no margins for duplication. Close cooperation with the entrusted entities is expected through regular meetings related to specific project implementation and in the context of the SRA-CSS. For illustration purposes, Annex C provides an indication of the expectations of the Entrusted Entities concerning new technologies, enlarged datasets and data processing paradigms.

Action B: Enlarging service outreach and improve response to user specific and regional needs

An action is sought for Horizon Europe call 2024 for Copernicus Security Service to test the feasibility of user-led regional services in support to law enforcement and maritime security applications, working closely with the Copernicus core services and complementing the ‘*Security*’ portfolio.

The objective of the action is to test a regional approach for maritime surveillance over a large EEZ. The proposed activity shall respond to the following requirements, or a subset:

On policy and strategic objectives:

- Account for EU, Regional and National policies in the maritime domain, in line with the priorities set by the EUMSS and the Blue Economy;
- Complement what is being offered by the Copernicus Maritime Surveillance Service while making full use of it as needed;
- Pursue a user-centric approach, by supporting users in their daily operations, promoting user-led initiatives and innovative solutions in cooperation with SMEs;
- Act as a regional facility for testing, demonstrating and benchmarking of innovative solutions and technologies, in particular those derived from new-space and from relevant Horizon Europe and National R&D programmes;
- Incorporate the concept of ‘rapid reaction’ mechanism for emergency response in the maritime domain, including pre-emptive tasking over risk zones, near-real time processing and delivery;
- Promote the development of EU technology and surveillance solutions aiming at European strategic autonomy;
- Promote dual-use services among a broad set of users;
- Prototype lean operational and technical deployment through innovative surveillance solutions and business models;

On communication and knowledge sharing:

- Promote a network on the exchange of best practices at EU level, for results applicable at Regional or National level;
- Organise of ad-hoc networking and awareness events aiming at user uptake, awareness and exchange of best practices, liaising also with training in promoted under the European Coast Guard functions training network;

- Promote and exchange know-how and cross-fertilisation across other regional initiatives such as the Copernicus Arctic Thematic hub;
- Whenever needed along EU policies, promote cooperation and exchange of best-practices at international level (*e.g.*, capacity building with neighbouring countries or critical strategic areas);

Ultimately, it is expected that this action will:

- Deliver a concept of operations and business model for deployment at regional level for 2027+.

Medium to Long-term (end of current MFF and next MFF) R&D priorities

R&D planning for 2026+ will follow the structured and inclusive methodology approach in the CSS-SRA. At the pace technology is evolving in this domain, detailing actions with a horizon exceeding 2-3 years would risk obsolescence and jeopardise innovation. Nevertheless, high-level orientations and funding requirement are indicated herein. Every year the CSS-SRA will be updated to reflect the lessons learned, map new technologies and initiatives, as well as update service and end-user needs based also on on-going R&D intermediate results.

High-level priorities for 2026+ were discussed at the CSS-SRA workshop held in December 2023 in Lisbon. It became clear that *New Space* start-ups will introduce additional capabilities, some possibly relevant to Security applications (pending actual operational specifications), such as hyperspectral, thermal, and emissions detections payloads. Member States also plan to introduce new capacities relevant to security applications. Italy (IRIDE) and Norway plan to launch new capacities as of 2026, with similar plans from Portugal (Atlantic constellation). The commercial sector is enhancing resolution, coverage, shortening revisit times and introducing new sensors such as radio frequency but also new EO services exploitation models. Copernicus will also have new relevant capacities, in particular the L-band SAR mission ROSE-L, promising for maritime surveillance and planned for 2027, and new hyperspectral capacities in CHIME, as from 2029.

Users indicated a strong interest in wider coverage, more flexible tasking to support short-notice requests, fusion of multi-sensor information, hyperspectral and thermal new capabilities, tip-and-cue capabilities but also a simplified access to historical data. Reinforcement of capabilities over the Arctic and Atlantic were indicated as priorities, as well as extended coverage for fisheries control. Monitoring of critical infrastructures, in particular over large unmonitored maritime domain areas is clearly a priority for EU security expressed by some users already working in the area, albeit more work would be needed.

Validation of those elements in operational theatres with the active participation of users will be crucial, before incorporation into the security services supply chain, through operational procurement. This should also take into account the need to improve the European strategic and economic autonomy.

As the discussions under CSS-SRA have shown, efficient coordination between actors is vital, emphasizing the need for mature R&D results validated in operational scenarios, not only from current Horizon projects but also from activities pursued by ESA and the Industry. Complementary should also be envisaged and defined upon programmatic definition, with ESA activities focussing mainly on technology aspects and Horizon Europe on the next step, service pre-validation through suitable operational testbeds led by the user community and with European and National strategic priorities in mind.

A long-term strategy will need to ensure continuity and coherency of actions, also to maximise the probability of exploitation of results. All R&D actions will need to be streamlined along a long-term vision and fed by the results of related precedent activities.

As such, long-term support to the effort initiated in CSS-SRA 2023 is expected. *Action B* in *HORIZON-CL4-2024-SPACE-01-36*, is a first attempt to respond to the need of operational testing, where technology can be demonstrated and possibly validated with a strong user involvement. Although short-term actions are focussed on maritime surveillance in response to the EU Maritime Security Strategy priorities revisited in 2023, its scope should be expanded to other security domains. The action will also help pooling know-how through the exchanges of lessons learned among parties, key to fostering user uptake and to capture and consolidate evolving requirements.

One of the major points indicated by users at the CSS-SRA Workshop in 2023 relates to the fitness-for-purpose of the services delivered. In addition to the limitations on the current space infrastructure, processing capacities must be improved. Today, validation of services delivery, such as dealing with false positives, but also the integration of additional information can be very time-consuming activities, which are dealt today by the final user. Quality control, automation, and data fusion are areas where R&D can play a key role.

Action A in *HORIZON-CL4-2024-SPACE-01-36* is a step forward in finding ways to improving the situation. Here a long-term sustainable action is needed, accounting for new sensors and evolution of technology related to processing capabilities (in ground and on-board) and new EO service models.

Coordination is a pre-condition to success. Financial resources are scarce, and Europe can't afford inefficiency caused by fragmentation and duplications. This is where the CSS-SRA process can play a key role, convening coordination meetings between stakeholders, on a bilateral or multilateral basis as needed, to tackle common problems and identify actionable roadmaps for concerted solutions.

Last but not least, as series of good practices should to be put in place for R&D activities, accounting for, namely, but not exclusively:

- Priority shall be given to activities driven by operational requirements;
- Promoting innovation and EU strategic and economic autonomy;
- Effective upstream screening of unnecessary duplications⁽¹⁰⁾;
- Screen activities that would better fit under service procurement. Use pre-commercial procurement funding, if and when needed;
- Foster information exchange among R&D activities, through appropriate communication actions;
- Involve Entrusted Entities in accompanying R&D activities relevant to the mandate of the service they are entrusted with, from inception to delivery, through adequate advisory mechanisms;
- IPR on the exploitation of results must be clear upstream, especially if an Entrusted Entity will be a partner in the project;
- Fully exploit the R&D work-packages foreseen in the Contribution Agreements related to the three Copernicus Security services, including technology spin-in;
- The direct involvement of a CSS Entrusted Entity in a Horizon Europe project should be clear and justified, taking fully into account their role as supplier of core services;

⁽¹⁰⁾ In particular proposals for development or demonstration of existing EU capabilities and services

- All the activities should deepen synergies with other Copernicus services, particularly with CEMS on technology needs, CEMS and CLMS on data exchange, and C3S on climate data for SESA.

The CSA-SRA will further detail the process, actions and progress of the elements above.

Considering what has been discussed among stakeholders, the tentative roadmap for 2026-27 will need to account for the following financing elements, pending yearly validation, prioritisation and refining of objectives through the CSS-SRA process:

- Sustainable finance of testing and validation of space data for security applications, with a strong user involvement and along EU strategic and economic autonomy objectives; build on the results of *Action B* and on the new technology meanwhile available. Timeline: start 2027 to avoid gaps (WP26), spanning 3 years. Relates to Cluster 4;
- Empowering end-users, promote data fusion for added-value services and the exchange of best practices; one or more actions (regional/thematic):. Timeline: Start 2017 (WP26) to avoid gaps, spanning 3 years. Build also on *Action B*. Relates to Cluster 4 and section 4.4 in the SRIA, and Cluster 3;
- Improve data processing capacities for security applications. This activity is closely related to similar needs from other services and should be pursued in synergy with other services. Customised for Security application needed. Relates to Horizontal activities action 5.4 in the SRIA, Cluster 4 and Cluster 6. Timeline: Start 2027-28 (WP 26-27) based on results from *Action A*. Resources for the Security part. Timeline: synchronised with related actions;
- Actions demonstrating complementary use and data fusion with other observation capacities and data – (eg. HAPS, UAVs, underwater observations, *etc.*). – Cluster 3. Timeline TBD.

Key R&D partnerships at EU and International level

Close cooperation is foreseen with CEMS (technology), CMEMS (marine data), CLMS (environmental compliance) and C3S (climate adaptation/human migration). Due to its nature, no partnerships are foreseen for R&D at international level.

4 Upstream and downstream elements

4.1 Space and ground segments - ESA

Upstream - Space segment

It is useful to recall the perimeter of upstream, space segment activities linked to Copernicus within ESA:

- Generic technologies for space systems that can be of interest to Copernicus are addressed in a dedicated EU/ESA coordination scheme. On ESA side, EOP representatives participate in the internal discussion, ensuring consistency and non-duplication of activities between different programmes;
- Mission implementation beyond phase A/B1 are in the Copernicus programme, hence depends on FFPA/CA activities and governance;
- Early phases of potential future missions for Copernicus Space component are fully funded by ESA as part of the FutureEO programme block 1, which sets the foundation of preparing future EO space systems addressing research needs (*e.g.* Earth Explorers), operational needs (*e.g.* preparation of Sentinel Next Generation or future meteorological programmes). This includes elaboration of Mission Requirement documents in interaction with users, phase O/A system studies, instrument predevelopment studies, and EO-specific technology activities.

It is therefore highly desirable to maintain a high level of coordination between the current programmatic frameworks as part of the SRIA for Copernicus. One such possible mechanism is to build on the Joint EC-ESA Earth System Science Initiative (ESSI) where the Research and Innovation Directorate of the European Commission (DG-RTD) and the Earth Observation Programmes Directorate of the European Space Agency (EOP) aim at establishing an effective alignment of selected scientific activities under their Horizon Europe and FutureEO programmes, in terms of goals, content, and planning in order to jointly advance Earth system science and its contribution to global challenges. DG RTD and ESA are cooperating on the basis of a common Scientific Agenda, including a set of common scientific priorities. As a response to the Scientific Agenda, a set of Flagship Actions is being defined to be implemented through a coordinated multi-year set of studies, projects and actions to be funded by both RTD and ESA following their own areas of expertise, competences and funding and programmatic mechanisms. Embedding Copernicus in such coordination would help identifying initial requirements for new or enhanced measurements of geophysical parameters, identify gaps or shortcomings in available or planned spaceborne remote sensing technics, and ensure gradual uptake and transitioning of EO related science into corresponding services.

With regards to upstream/space segment activities, visibility on respective workplans and activity reports (*e.g.* QSR on ESA side) should be maintained, and regular exchanges on strategic perspectives should be actively pursued in order to maintain a profitable coordination.

R&D in support to products and data quality

Context and rationale

A major asset of the Copernicus Space Component lies in the quality of the Sentinel core Level-1 and Level-2 data products. In this respect, ESA in cooperation with partner agencies, conducts the appropriate R&D activities for Sentinel missions to maintain high-quality data and services, enabled not only by the excellence of their technologies but also by the Space Component of on-ground efforts in terms of calibration, validation, and processing algorithms performance. Those activities are fundamental to

preserve data accuracy and stability as well as to maintain the necessary level of confidence required for a wide spectrum of operational and science domains (*e.g.*, applications for Climate). A sustained and focused effort on R&D should be enhanced to support evolutions of core data products and guarantee the end-to-end performance of the overall Copernicus Programme. There is a clear need to further invest in that domain and coordinate ESA-led activities with SRIA to get the best out of the Copernicus Programme and facilitate that way, the interoperability and synergy of the Sentinels first generation missions and Copernicus Contributing missions with other EO satellites, including the future Copernicus Expansion Missions (CEM) and their integration into Copernicus services.

R&D and short-term improvements of Sentinel core data products

As a consensus and ESA lesson-learned in operating Sentinel-1, 2, 3 (land part) and 5-P missions, there is a continuous need to further reinforce R&D and Cal/Val activities to support improvements of level-1 and level-2 operational processing algorithms and derive traceable uncertainty parameters. Those activities managed by ESA are instrumental to guarantee the quality of Sentinel core data products which are now used as the reference in Cal/Val for NewSpace and Copernicus Contributing Missions. Enhancing the data quality of Sentinel missions and characterizing uncertainty parameters all along their lifetime is also necessary to ensure their proper exploitation by Copernicus services for higher level processing stages, data assimilation into operational models, downstream applications as well as in support to science projects funded under the Horizon Europe framework.

Aligned with the fast-evolving SAR Analysis Ready Data landscape, one of short-term R&D priorities for Sentinel-1 is the definition of algorithms and processing flows which shall allow for a harmonized family of Analysis Ready Data (ARD) ranging from detected products to ARD which explore the complex nature of the SAR data (*e.g.*, geocoded Single-Look Complex Data or Interferometric Data). On the geophysical products (Ocean Level-2), priorities include the expansion of the product family such as wave parameters retrieval. This should allow for a significant extension of the coverage of the product, further promoting data assimilation in weather models. For the ocean wind fields, the focus is on the improvement of the inversion with the addition of new constraints to remove of the a-priori wind information from ancillary data, which should extend the usability of the SAR retrieved product.

For Sentinel-2, a line of short-term R&D activities is devoted to the improvement of atmospheric correction and the very important cloud screening algorithms generating Level-2A operational products. Despite the great success in improving these algorithms, these domains are still maturing within the scientific community, with international efforts exchanging in the framework of CEOS ACIX and CMIX activities. A second line of R&D activities focuses on the improvement of the pilot products Level-2H (harmonizing Sentinel-2 and other missions such as Landsat and PRISMA) and Level-2F (adding a fusion step to Level-2F product to bring spatial resolution of all input missions to the one of Sentinel-2). R&D will focus on bandpass adaptation, bands inter-calibration and BRDF (Bi-directional Reflectance Distribution Function) corrections. It would be important to define a common reference or standard in this context.

Additional R&D are also recommended by the Copernicus Services and the user community in the field of Sentinel-3 altimetry, especially over complex areas over the sea ice and ice sheet margins where the potential of core level-1 and level-2 data products is not fully exploited. Such activities are fundamental not only for Climate, to narrow the gap between the Sentinel-3 Altimetry mission, the ESA CryoSat and CRISTAL but also for a better integration into operational numerical models. The Sentinel-3 Altimetry

mission is also providing a wealth of new and meaningful data over lakes and rivers, that need to be further exploited to better understand the water cycle in a changing world. On the optical side, the Sentinel-3 SYnergy Surface Reflectance product is undergoing an evolution towards becoming an Analysis Ready Data (ARD) product compatible with the ARD requirements in the medium term.

Regarding atmospheric products from Sentinel-5 Precursor (S-5P), the ATM MPC (Atmospheric Mission Performance Cluster) is responsible to carry out the data Quality Control and Validation on operational level. This scheme is in place since 2017, since the launch of the Copernicus mission (the ATM MPC was then called SSP MPC).

Long term R&D and preparation of the Copernicus Expansion Missions

Despite the core sentinel products being already mature, several R&D are running or planned by ESA in cooperation with EUMETSAT to further improve their quality on longer term and to prepare the development operational processing algorithms for Copernicus Expansion Missions (CEM).

Here below is a non-exhaustive list of priorities already identified by ESA.

The planned approach for Sentinel-1 is to develop algorithms which can be also extended to ROSE-L CEM and favour multisensory and multi-mission applications. ROSE-L Level-2 product development will focus on the retrieval of forest and ocean variables. This shall allow both for continuity of Sentinel-1 geophysical products and for the synergy with products and applications from other ESA and non-European upcoming missions (*e.g.*, BIOMASS and NISAR), promoting the development of multi-mission and multi-frequency applications.

On the optical side, a long-term line of sentinel-2 R&D regards the improvement of the aquatic reflectance product. This product is currently under development based on the operational algorithms used by the Copernicus Services, and further improvement of the algorithms (*e.g.*, in terms of glint removal, white caps removal or atmospheric adjacency effects correction) is expected. Also, the preparation of LSTM activities is ongoing, with the development of the LSTM L2 Processor and Product aiming towards developing a product with different thematic layers for Land, -Ice and -Sea surface temperature as well as land and aquatic reflectance layers. For CHIME, L2 activities will focus on the Bottom-Of-Atmosphere (BOA) reflectance retrieval fulfilling the CEOS ARD surface and aquatic reflectance specifications. Additionally, a range of higher-level high priority products related to canopy & leaf level and soil & mineralogy will be prototyped.

Beyond Sentinel-3 based R&D to improve current altimetry processing methods for polar and inland water, the upcoming passive and active microwave CEM will also require dedicated R&D efforts. For CRISTAL, the main challenge will consist in the characterization of dual band acquisitions over the ice. This implies studying and understanding the different scattering from the Ka and Ku band as function of the snow layer and the different resolution cells. The operational algorithms shall be defined to combine and harmonize the measurements at the Ka and Ku bands allowing as final aim to infer the snow characterization. Exploiting the on-board radiometer acquisitions to characterize/classify the sea ice overflow will be also a challenge to improve the retrieval of sea ice variables. For CIMR, significant effort will be spent to define consistent and physically based joint retrieval algorithms for the geophysical variables in each thematic domain (*i.e.* Polar regions and land) to leverage the full capabilities of the on-board instruments.

On longer term, effort shall be made to define level-1 and level-2 retrieval algorithms exploiting co-located acquisitions from different sensors, especially the ones relevant to the polar regions (i.e. CRISTAL, CIMR and ROSE-L). This should foster synergies among the different CEMs to fully exploit the capabilities of the Copernicus constellation and address the EU Arctic policy. As an example, CRISTAL altimetry measurements could be used in synergy with other high-resolution data such as ROSE-L imagery to jointly estimate the extent and elevation of icebergs and in turn to estimate the iceberg volume.

Perspectives for Copernicus mentioned in the Copernicus Space Component Long Term Scenario include gravity field measurements, in particular quantum gravity prospects. On a shorter timescale, continuity and enhancement of current capabilities are implemented through the US/Europe MAGIC constellation concept. The NGGM mission is the ESA-led component and is designed to provide better performance in several systems elements such as satellite-to-satellite tracking, improved instrumentation, orbit control: these enhancements are necessary to match the increased sensitivity expected from quantum gravimetry. Preparing for a potential sustained European gravity field measurement capacity in Copernicus requires to keep a strong synergy between NGGM development on ESA side and other quantum-related R&D studies, Quantum Space Gravity pathfinder mission developed through Horizon Europe, and other potential preparatory studies to strengthen user requirements, operational services.

FRM and Cal/Val activities

A priority of ESA is the harmonization and standardization of calibration and validation (Cal/Val) procedures as well as the fostering of multi-mission intercomparison exercises. The Cal/Val activities ideally rely on Fiducial Reference Measurement (FRM) data as reference and represent a key element for the performance of EO missions to give confidence to the end users in the quality of the data products. Many ESA initiatives have been conducted in this domain such as FRM4SOC, FRM4SST, FRM4ALT, FRM4SAR, FRM4GHG, ... This effort is to be pursued for the Copernicus Expansion Missions as a preparatory activity for all new geophysical variables and associated uncertainty parameters. Those activities include the definition, development, and data provision of FRM in support to the validation of the Copernicus core products and in complement to existing in-situ networks. Preparatory activity to set-up FRM for the future Copernicus Expansion Missions (CIMR, CRISTAL, LSTM, ROSE-L, CHIME) will be also fundamental to guaranty the availability of a ‘trustable’ reference against which performance of the core products can be anchored or assessed against. Moreover, the plan is to foster the “supersite” concept, promoting the establishment of FRM in key areas covered by different Sentinel and Copernicus Contributing missions.

The expanding SAR landscape driven both by Space Agencies as well as the NewSpace sector together with the increase of freely available SAR data to the science community highlight the need of a sound and standardized Cal/Val approach. This is not only to ensure the high quality of data processed by ESA, but also to promote the development of multi-sensor and multi-mission applications and to promote the usability of NewSpace data. Accordingly, a priority is on the harmonization and standardization of Cal/Val procedures as well as the fostering of multi-mission intercomparison exercises. The Cal/Val approach ideally relies on FRM data as reference. This is indeed in place for the Cal/Val of Level-1 Sentinel-1 data, which mainly rely on few high-quality artificial targets. Nevertheless, there is the growing need to increase the calibration networks as well as to foster the open sharing of standard information about calibration targets. This is crucial to ensure the quality of S1 and ROSE-L Cal/Val approaches and, consequently of the data. Moreover, this is fundamental to achieve harmonized calibration between different missions,

promoting synergy. The use of FRM for Level-2 SAR Cal/Val is less consolidated w.r.t the Level-1 case. This is due to the lacking availability of global forest and ocean FRM collocated with the SAR measurements. Hence, a long-term priority is the refinement of the definition and the collection of forest and ocean FRM.

A new activity endorsed by CEOS has started, aiming at developing a network-based Thermal Infra-red top-of-atmosphere calibration capacity for thermal sensors of ~100m or higher spatial resolution. The network prototyping is an on-going activity with the objective to have several operational sites in the longer term providing upscaled ToA brightness temperature measurements measured on ground for radiometric calibration and validation similar to the RadcalNet network for visible sensors.

It is also of key importance to guarantee high quality, long-term and consistent data product over the polar regions especially to address the EU Arctic policy. In particular, the need to bridge the gap with CRISTAL is a key priority for the polar science community to guarantee the continuity of Ku-band radar measurement in polar regions above 82 deg. This could be achieved *e.g.* with systematic airborne/in-situ campaigns in polar and in building on 20+ year experience gained in this field with ESA-led CryoVEx/CryoVal campaigns. Deploying and maintaining on the long-term sea ice and land ice FRM sensors will be also crucial to support a consistent validation of the products during the Mission lifetime as well as to allow for efficient intercalibration among different Missions operating in the polar regions and exploiting different instruments. The elaboration of future polar and inland FRM for CRISTAL will build on the ESA St3TART (*Sentinel-3 Topography mission Assessment through Reference Techniques*) which is an EU and ESA funded project aiming at generalising the concept of FRM for validation of the Copernicus Sentinel-3 STM over inland waters, sea ice and land ice.

FRMs are also the most important elements today to carry out the validation of the Level-2 data Sentinel-5P products, besides existing Networks (*e.g.* NDACC Network for the Detection of Atmospheric Composition Change). Several atmospheric FRM programmes have been developed in cooperation between ESA and partner agencies in the past 7 years, including the Pandonia Global Network for which ESA supports operational and R&D activities for further development of the instrument and algorithms. FRM4GHG is the R&D activity related to GHG data products as ground-based data sets. COCCON has been also developed through ESA funding on a small scale. The data sets are important in view of the upcoming CO2M mission and, in close collaboration with EUMETSAT, R&D activities are coordinated, as different requirements need to be fulfilled. FRM4DOAS is another R&D programme used in the context of UV-VIS S5P data set validation. The operational implementation is still standing out but expected to be activated in early 2024. The future operations of TCCON FAST will allow to deliver TCCON data within at least 3 months upon acquisition. Data harmonization aspects are the focus in this activity, funded through the EU. The ESA atmospheric Data Center (EVDC) is a data base for searching and accessing correlative data for calibration and validation of atmospheric satellite products. It provides access to ground based, aircraft and balloon Cal/Val files and FRM. It also provides access to selected atmospheric Satellite data (Sentinel 5P, Aeolus and MIPAS, EarthCARE in the future) and provides tools to perform validation tasks using the match up data base and several more functions.

ESA activity called Cal/Val Park is currently in an R&D phase; to the creation of an innovative Cal/Val site, developing ideas on the potential best design, targets and instrumentation to be installed. After its future construction, the Park will keep a playground part dedicated to evolution of innovative Cal/Val methods and ideas, in a sort of continuous R&D throughout the whole lifetime of the Park. The main target is to eventually serve the Copernicus Contributing Missions, in addition to the Copernicus Sentinel-2 and CHIME missions, helping to improve their Data Quality aspects as a whole. A Key partnership is already

defined with ASI and potential involvement of other key actors will be investigated. R&D activities are also planned with regard to the creation of both a Radiometric and Geometric Match Up Databases, with particular regard to helping the improvement of Copernicus Contributing Missions Data Quality & Cal/Val, somehow following up on the achievement obtained by the Copernicus Sentinel-2 GRI.

ESA EO Science for Society

ESA science and R&D activities (EO Science for Society) aims at maximising the impact of European EO assets in terms of new methods, novel products, advanced Earth system science results and innovative applications and services. This programme is based on a continuous consultation process with the community (scientist, users, industry) and include dedicated elements to support the exploitation of the current Sentinels in terms of science and novel applications but also a new Sentinel User Preparation element aiming at preparing for a fast uptake of Sentinel next generation and expansion missions by the community and maximizing industrial competitiveness.

ESA R&D activities may represent an important development pillar for Copernicus, its services and evolution. Already, several ESA funded developments have been transferred to both the Copernicus mission operations and to Copernicus Services. ESA R&D activities should be considered as a key contributor not only to develop capacity in Member States but also to contribute to realise the Copernicus R&I Strategy and foster the uptake of novel developments by the Copernicus Services and their evolution. Copernicus SRIA can benefit from results of the R&D activities undertaken by ESA in response to recommendations collected from a continuous and wide consultation process with scientists and industry. At the same time, relevant priorities identified by the SRIA should be addressed by ESA through dedicated R&D projects with the community, especially for appropriate fundamental EO research, early capability developments and EO science/ applications aspects.

ESA Climate Space

From 2023 to 2029, up to and beyond the 2nd UNFCCC Global Stocktake, ESA's CLIMATE-SPACE programme will support Member States by expanding its Essential Climate Variable portfolio which will further be transitioned to several Copernicus services, support progression towards UNFCCC Paris Agreement goals; provide pre-operational R&D for operational climate services, including the Copernicus services, enhance the link between the climate observation and modelling community, and conduct cross-ECV and Tipping Points studies. CLIMATE-SPACE builds on the success of the Climate Change Initiative, running for more than a decade, having established, through ESA's Climate Office, links to all stakeholders in the international climate network, including UNFCCC, IPCC, GCOS and WCRP. ESA's Climate Office hosts the WCRP Coupled Model Intercomparison Project Office (CMIP) since March 2022.

ESA EO Digital Twin Earth

Latest advances in EO science and research activities are opening the door to the development of novel EO-based Digital Twin Earth (DTE) that may contribute and maximise the impact of EO in the design and implementation of future Digital Twins ecosystems. ESA DTE will develop and demonstrate, up to a pre-operational level, a set of Digital Twin Components as EO-based replicas of a physical dynamic process in the real world and thus of our environment and their interactions with human activities and ecosystems with a strong focus on valorising the role of EO capabilities. ESA EO Digital Twin Components will target interactive dynamic reconstruction, and simulations of the Earth system and its interactions with human activities potentially under different conditions or scenarios.

ESA Φ-lab

Exciting NewSpace and scientific opportunities are opening up for the latest generation of digital entrepreneurs and data scientists, driven by a growing volume of increasingly rich Earth observation (EO) data, growing computational power in space and the untapped potential of game-changing technologies such as artificial intelligence (AI) (for example under the AI4EO programme of ESA) and Quantum and Edge Computing. ESA Φ-lab provides an open, dynamic and fast-moving innovation environment where talented people can develop transformative ideas – from blue-sky thinking through to entering the market – that will accelerate future EO by delivering enhanced capabilities and strengthening Europe’s resilience and world-leading competitiveness. Copernicus shall leverage such potential to transform its offer to address societal challenges.

4.2 Space and ground segments - EUMETSAT

Vision and mission

EUMETSAT’s mandate encompasses the delivery of global Earth Observation data relevant for weather, atmospheric composition, marine applications and climate monitoring, which are time-critical, operational and require long-term stable data flows. EUMETSAT generates single-sensor products as well as synergistic products between the EUMETSAT operated Copernicus Sentinels, Europe’s meteorological missions, and relevant third-party data from international space partners. This exhaustive pallet of products primarily focuses on addressing specific end-user needs, by using leading-edge scientific and computational developments, thus ensuring the highest quality, and fit for purpose product pallet for the end-users.

Main portfolio and expected evolution

EUMETSAT operates the European geostationary (Meteosat) and low Earth orbit (Metop) meteorological satellites as well as the Copernicus Sentinel-3, Jason-3, and Sentinel-6 missions, the latter on behalf of the European Commission. These missions generate a stable portfolio of a variety of products on weather, atmosphere and oceans, and those are constantly being improved and expanded with new products. In the next 2-5 years, EUMETSAT will operate the Meteosat Third Generation (with Sentinel-4 onboard), and Metop Second Generation (with Sentinel-5 on board) and CO2M Copernicus missions. This will make products on atmospheric composition (*e.g.*, ozone, NO₂, SO₂, aerosols, *etc.*) and greenhouse gases (Methane, CO₂, N₂O) operationally available in near real time for use by the Copernicus Atmosphere Monitoring Service (CAMS) and other users. In addition, the capabilities of the next generation EUMETSAT missions, combined with the capabilities of the new Copernicus missions, will allow for the developments of innovative product families such as for coastal zones, land cover products and surface properties, the cryosphere, and at high-latitudes/Polar zones, among many.

Target policies, customers, users

EUMETSAT’s data services primarily targets operational user communities, mandated to monitor the implementation of National and European policies related to meteorology, high impact weather and disaster reduction, air pollution and air quality, aquatic biodiversity, marine water quality, the arctic, and climate monitoring, amongst others. The Copernicus services are key users that require the data as input for their models and the generation of added value products. Additionally, the data services support applications on operational oceanography, disaster management, climate and atmosphere monitoring provided by governmental services, international institutions and the private sector, as such feeding into a wide variety of economic sectors such as transport, energy, tourism, health and agriculture.

Strengths, weaknesses, opportunities and threats

Europe's meteorological missions in combination with the Copernicus Sentinels, provide a quantity of high-quality data that is unprecedented globally, putting Europe at the forefront in Earth Observation. Furthermore, EUMETSAT is putting in place innovative state-of-the-art new generation of meteorological missions (MTG, EPS-SG, EPS-Aeolus, EPS-Sterna), that will provide opportunities for strengthening Copernicus services, as well as weather and climate monitoring and modelling. This is a major strength and provides significant opportunities for European actors, private and institutional, and ensures Europe's autonomy in generating environmental data. There are however four main weaknesses with related threats that are required to be addressed. Firstly, the European space assets are currently under-exploited in certain application areas, which could be addressed through the generation of multi-sensor or multi-platform synergistic products and cross-sensor calibrated long-term time series spanning various Copernicus services elements. Second, there is a strong dependency on non-European in-situ infrastructures to ensure availability of validation data globally to monitor and maintain the data quality of the European space assets. Third, gaps in radiative transfer models need to be addressed as this is fundamental to fully exploit the new-generation satellite sensor data. Last, the potential of AI/ML techniques for EO is to be further explored for large scale EO data monitoring and new product developments, which requires improvements in the seamless availability of European cloud-based platforms, allowing to manage/exploit the data volumes and providing flexible means for the development of AI/ML and EO-based operational applications.

Short-term (2025-2027) R&D priorities

R&D priorities are centred on the above-identified opportunities, weaknesses and threats, with both short-term and long-term elements:

- **Generation of synergistic products from Copernicus and non-Copernicus missions:** The MTG and EPS-SG are operational programs that each guarantee more than 20 years of observations. The satellites will fly highly innovative instruments, including the new optical imagers FCI and Metimage, the next hyperspectral infrared sounders IRS and IASI-NG, advanced microwave imagers MWI and ICI, CO2M, and the Sentinel-4 and -5 Copernicus missions. There will be a need for investing in R&D activities to get the most out of these new data, and to create innovation through their use, alone or in combination with other data, offering unique opportunities for developing innovative multi-sensor products for the monitoring of the atmosphere, natural disasters (*e.g.* wildfire monitoring), the oceans and climate change. In addition, for the preparation of the Next Generation and Copernicus expansion missions, R&D activities are needed, building on precursor missions such as Sentinel-5P, Microcarb and the NASA-CNES SWOT and other missions from international partners such as NOAA and NASA. Finally, integration and synergistic products between EUMETSAT and Copernicus missions and with NewSpace satellites in a hybrid constellation will enhance the observations delivered by Copernicus. EUMETSAT and ESA are also planning for the launch of new operational mission EPS-Aeolus and EPS-Sterna that will be critical for weather modelling and climate (notably in support to strategic arctic observations);
- **Improvement of Sentinel-3 and Sentinel-6 products:** Activities focus on both lower-level product improvements and evolutions (such as the improvement of cloud/snow detection, atmospheric corrections), the development of new products (*e.g.*, sea ice surface temperature, water vapour/wet tropospheric corrections, aerosol layer height) and adaptation of existing products for specific environments (arctic- leads zones, coast);
- **Address gaps in radiative transfer models to realise the innovation potential of new missions:** new generation Copernicus and contributing missions have much improved performance (radiometric, resolution, spectral) compared to the current suite. The advancements in radiative transfer

modelling has to keep pace ensuring a full utilization of the data from these missions. R&D is needed in areas of spectroscopy, radiative properties of clouds, aerosols, precipitation and complex surfaces, and polarised radiation to instigate development of accurate and fast radiative transfer models to provide a major step forward in improving the quality of the products that support the downstream applications of the data from these missions;

- **Availability of Fiducial Reference Measurements (FRM) data and standardization of calibration/validation approaches:** Standardization of calibration/validation activities are needed for ensuring high quality and accurate Copernicus and EUMETSAT satellite mission data. Here, EUMETSAT and Copernicus can provide key inputs for the development of internationally recognized standards. Activities focus on deriving standards within a global/international context, for example, in continuity of activities started under FRM4SOC for traceable ground-based measurements of ocean colour parameters. Additionally, the Ocean Colour system for Vicarious Calibration, to be established under Copernicus, and TRUSTED for Sea Surface Temperature, will offer additional opportunities for the ocean monitoring community. As regards CO2M, R&D activities are needed for the standardization and improvement of ground-based measurements, within global networks/initiatives such as TCCON, COCCON, and the Pandora Global Network (PGN). FRM activities could also be reinforced by targeted field campaigns, be they air, sea and ground, and using European assets and research infrastructures, that would support the understanding of geophysical processes (and uncertainties in satellite products) leading to improved EO products;
- **Usage of AI/ML techniques for the development of cloud-based pathfinder applications:** The amount of data generated by the Copernicus missions operated by EUMETSAT will increase to more than 10Pb on a yearly basis. This will require innovative ways to handle the storage and retrieval of relevant information, as well as for the performance monitoring of the instruments data flows. Particularly interoperability and standardization of interfaces and formats will be required to fully benefit from the data generated. Applicability and usefulness of AI/ML techniques will need to be investigated to augment platforms such as Copernicus WEkEO for developing operational applications in the domain of atmosphere, environment, oceans and climate monitoring and modelling, also leveraging innovations from the DestinE Data Lake;
- **Data Curation – AI ready data:** to accelerate the uptake of AI/ML, there is a need to generate ad hoc datasets by combining existing observations with a series of data curation actions. These actions may include remapping data into reference grids (for example, if you want to target nowcasting, you would want your MSG/MTG rapid scan data on the Opera grid); anomaly detection and mitigation; metadata labelling. In addition to the rapid provision of such datasets for swift development, the maintenance and continuous improvement of forecasting systems in their operational environment require a strategy to consistently develop such datasets for re-training and ongoing enhancement. This principle applies to nowcasting, NWP, and seasonal- or decadal-climate forecasting.

Long-term (next MFF) R&D priorities

- Upcoming Copernicus missions will provide significant opportunities for the monitoring of oceans, atmosphere and greenhouse gases. This includes not only the Copernicus Expansion missions (CO2M, CIMR, CRISTAL) but also of the Next Generation Missions Sentinel-3 NG OPT/TOPO. The latter ones will deploy potentially novel technologies for which significant research will be necessary to fully exploit its benefits.

- The Copernicus altimetry missions, currently operational and to be deployed (CRISTAL, S3 NG TOPO, Sentinel-3 C/D), in combination with other non-Copernicus missions, will offer opportunities for the development of a global integrated system for the monitoring of sea level, waves and winds.
- FRM for ocean, atmosphere, and greenhouse gas monitoring will need to be sustained on a long-term basis, globally. FRM in-situ data, together with the appropriate instruments and validation techniques, will need to continuously evolve and further automated to improve cost/benefit of those activities.
- AI/ML driven research and development for EO new products, high data volume monitoring and diagnostics, using expanding cloud processing capabilities, will lead to improved European capabilities for end-users.
- Operational research on generative AI empowering users to explore Earth Observation data and make informed decisions for weather and climate operations. Additionally, auto-generative AI will facilitate interfacing with complex data sets, guiding user requests, and rapidly analyzing extensive data or generating related code.
- Acknowledging that Copernicus Sentinel data gradually are generating long-term climate data records (CDRs) it is important to have uncertainty-quantified time series that allow confidence statements for their usage in policy and other applications. This requires application of metrological techniques to satellite multi-platform and multi-sensor data to generate cross-calibrated, uncertainty-quantified sensor observations and Essential Climate Variables (ECV) data records. Research is needed to implement these techniques for the Copernicus data records, including also uncertainty estimates in data assimilation and machine learning applications.

Key R&D partnerships at EU and International level

Within the EU, there are ongoing cooperation activities with universities, the European Earth observation industry, the Copernicus Entrusted Entities and space agencies such as CNES, DLR, ESA. The Satellite Application Facilities are dedicated centres of excellence for processing satellite data and form an integral part of the distributed EUMETSAT Application Ground Segment. Key partnerships exist with NOAA, NASA, Japan, Canada, South Korea in the context of weather, atmosphere and marine earth observation satellites. Furthermore, a close interaction is pursued with international networks such as IOCCG, GHRSSST, OSTST, TCCON, PGN, AERONET, NDACC, and the European RFIs (IGOS, ACTRIS). EUMETSAT is a key actor in the CEOS and CGMS working groups, in particular the Joint CEOS-CGMS Working Group Climate, CEOS Virtual Constellations (CEOS – VC) and the CGMS Global Space-based Inter-Calibration System GSICS.

4.3 In-situ networks– - EEA

Vision and mission

The Copernicus Entrusted Entities are responsible for securing the in-situ resources that they need for their day-to-day activities. But given the complexity of the in-situ landscape and the diversity of requirements, a cross-cutting coordination is needed to prioritise actions in the in-situ domain, ensure that cross-cutting needs are well understood and handled in an efficient way, and tackle issues which cannot be dealt with satisfactorily by the Entrusted Entities individually. The European Environment Agency (EEA) plays a key coordination role at European level on those in-situ horizontal issues, which benefit all services.

Main portfolio and expected evolution

The Copernicus Entrusted Entities have their own activities and evolution plans when it comes to in-situ data based on their own requirements. In the context of EEA's cross-cutting coordination activities, the

main tasks are structured around four areas: the maintenance and update of the overview of the state of play of in-situ data for Copernicus services; the provision of cross-cutting access to in-situ data to Copernicus services; the management of partnerships with key data providers; and the provision of advice on in-situ matters to the European Commission and Copernicus Entrusted Entities.

Target policies, customers, users

The activities related to the access to in-situ data should meet specific requirements expressed by the Entrusted Entities and Commission, and it typically involves engagement with in-situ data providers, including pan-European networks, national institutions, research infrastructures and programmes, crowdsourcing initiatives, and private sector.

Strengths, weaknesses, opportunities, and threats

Strengths: There is a strong commitment from the Entrusted Entities and comprehensive and up-to-date knowledge about in-situ data requirements across the programme, as well as a continuously monitored platform that allows services to get access to national and regional geospatial data (CORDA). Good spirit of collaboration and strong engagement with key in-situ data networks and data providers, such as EuroGeographics, EUMETNET, EuroGOOS, EOOS, EMODNET, etc.

Weaknesses: The in-situ data landscape is extremely fragmented and complex. Restricted and/or delayed access to in-situ data hinders the full potential of their use in Copernicus programme. Sustainability of key in-situ networks that build upon research projects with short-term funding is a major risk for Copernicus services.

Opportunities: the implementation of the Open Data Directive and its High Value Datasets and a general trend towards open data should result in an important increase in the offer of authoritative in-situ data from European public data providers. The interest of collaborating across Copernicus Entrusted Entities in the context of AI may also provide leverage in sharing further in-situ data for AI algorithms training and validation.

Threats: in a period of increase inflation and higher costs, funding of in-situ activities may be impacted, including the sustainability of key in-situ networks.

Short-term (2025-2027) R&D priorities

R&D priorities associated with coordination of Copernicus' access to in-situ data are closely linked to key cross-cutting in-situ data requirements expressed by the Copernicus Entrusted Entities. These requirements are typically related to data availability and sustainability. Therefore, priorities are centred around elements such as:

- optimized use of observations delivered via IoT and other alternative sources, such as drones and other autonomous systems, webcams, citizen science and their efficient combination with conventional observing systems;
- use of AI to maximize in-situ data content and quality and develop AI-ready data sets (collection, generation, annotation/tagging);
- development and test of sustainable, stable, and manageable observing networks in data-sparse regions, including capacity building and governance aspects. There is also a need for research activities leading to the preparation of new consolidated datasets such as an upgraded Global Precipitation Dataset integrating satellite, in-situ and reanalysis data.

Long-term (next MFF) R&D priorities

Long-term R&D priorities associated with coordination of Copernicus' access to in-situ data concern topics such as:

- the development and use of new, low-cost instruments/sensors *e.g.*, in the domain of air quality, greenhouse gases, and soil moisture / water balance;
- specific data rescue activities and related developments;
- Cal/Val methodologies and Fiducial Reference Measurements (FRM) supporting the optimal use of data from Copernicus Expansion Missions, including in data poor remote and high latitude regions.

Key R&D partnerships at EU and International level

Key partnerships at EU level are already established via partnership agreements with EUMETNET, EuroGOOS/EOOS, EuroGeographics, EuroGEOsurveys. EEA also builds on its European Environment Information and Observation Network (Eionet) and eight centres of thematic expertise contracted by the EEA, called European Topic Centres (ETCs). Future partnerships can also be established with the European Research Infrastructures. Copernicus international arrangements can also provide a framework for further collaboration.

4.4 Market uptake– - EUSPA

Vision and mission

As per the Space Regulation, EUSPA’s mandate covers specifically the Copernicus market uptake addressing ‘other users’, the research aspects of which are addressed here.

EUSPA’s core mission is to implement the EU Space Programme and to provide reliable, safe and secure space-related services, maximising their socio-economic benefits for European society and business. By fostering the development of innovative and competitive upstream and downstream sectors and engaging with the entire EU Space community, EUSPA is driving innovation-based growth in the European economy and contributing to the safety of EU citizens and the security of the Union and its Member States, while at the same time reinforcing the EU’s strategic autonomy.

Vis-à-vis the Copernicus Strategic Research and Innovation Agenda, EUSPA structures its downstream priorities leveraging the market segmentation approach, to collect requirements and align them with its strategic objectives, drivers and orientations. The current input is based on “**EUSPA’s R&I downstream strategy for 2025-2027 and beyond**”⁽¹¹⁾ as defined and shared with the European Commission.

Main portfolio and expected evolution

EUSPA’s current R&I portfolio spans transversally across the EU Space Programme components, also exploiting synergies among them; a unique element that brings additional value to the space downstream ecosystem. EUSPA is currently acting on three main axes of work, which correspond to political priorities of the European Union, underpinning them with EU Space-powered applications that can foster the commercialization and adoption of EU Space data and services: i) the green transition with applications that support the EU Green Deal, ii) applications that exploit EU Space Programme data streams and evolve Europe’s digitalization efforts, iii) applications that reinforce resilience to security threats, safety hazards and crisis situations.

⁽¹¹⁾ EUSPA R&I downstream strategy for 2025-2027 and beyond, EUSPA-MDI-RAD-1LM-A23589, approved 02 April 2023.

The EUSPA R&I Copernicus objectives are to:

- Support the evolution of Copernicus and promote the synergies with other components of the Space Programme;
- Contribute to the EU priorities, including the EU Space Strategy for Security and Defense as appropriate,
- Foster cutting-edge innovation and growth to ensure that the European space industry continues to deliver value to the end-users and customers, efficiently exploiting Copernicus data and services.

EUSPA shall continue to develop synergies of Copernicus with Galileo and the secure SatCom component (*i.e.*, GOVSATCOM and IRIS² in the future) and support industry in the valorisation of new Copernicus missions. In this line, the Agency shall also seek to increase the penetration and reach of Copernicus, through international cooperation, supporting its internationalization and adoption from non-European users globally.

Target policies, customers, users

EUSPA defines strategic orientation for the R&I downstream priorities, based on inputs gathered from stakeholders via consultation as the User Consultation Platform (UCP), workshops' surveys and built on market intelligence (bottom-up). Its "customers and users" are the ones acting in all the market segments that EUSPA defines in its market segmentation. This approach enables very specific market-oriented inputs to be collected to shape future R&I priorities. The policy priorities foreseen to be supported are presented inside the following macro-segments for better granularity (non-exhaustive):

- Mobility and Space (Aviation & Drones, Road & Automotive, Maritime & Inland waterways, Rail, Space): EU Reducing emissions from aviation, Regulation for autonomous vehicles;
- Consumer solutions (Tourism, Health, Insurance & Finance): EU Strategy for Sustainable Tourism;
- Energy & Environment (Energy, Biodiversity, Forestry, Agriculture, Raw Materials, Climate Services, Environmental Monitoring, Fisheries & Aquaculture): EU Green Deal, ESG-related legislation in the EU, Illegal, Unreported and Unregulated Fishing (IUU) control, Critical Raw Materials Act, Biodiversity Strategy 2030, EU regulation for deforestation, Regulation on Land Use, Land Use Change and Forestry (LULUCF), *etc.*;
- Governmental (EMHA, Security & Surveillance, Infrastructures): EU Space Strategy for Security and Defense, EU Cybersecurity Act, Disaster Resilience Goals, *etc.*

Strengths, weaknesses, opportunities and threats

Copernicus is a unique and strong Earth Observation Programme that shall evolve, given the rapid changes existing in the space ecosystem today. According to EUSPA's EO & GNSS market report², European companies accounted for half of the global Analysis, Insights & Decision Support market in 2019. During the last 10 years, the NewSpace paradigm led to the diversification and proliferation of space data sources, opening opportunities for expanding the Copernicus offer. Responsiveness, assurance and security of the delivered services is critical for many applications. Threats are largely triggered by the current geopolitical situation and will impact the way and velocity in which the Union performs 'research-to-operations', to be reinforced. Copernicus will have to evolve in this ecosystem and continue to offer state-of-the-art services that shall build on analysis-ready data to enable sustainable downstream linkages with the market, possibly co-existing with commercial products and services and/or incorporating them in its portfolio. EUSPA will support the evolution of the Copernicus downstream development, embracing new capabilities of the future Sentinel missions, leveraging and underpinning service evolution developments, digital services (*e.g.*, HPC, Cloud) and Artificial Intelligence.

Short-term (2025-2027) R&D priorities

For the period 2025-2027, the main EUSPA Copernicus R&I priorities are (reported per market segment – not prioritized):

- Enable advanced EO applications that increase the resilience of urban areas, infrastructures and societies and support the efficiency of **mobility** infrastructure management, including maintenance and monitoring of *e.g.*, road and rail networks. Support for cost-efficient, optimised and faster bathymetry extraction for **coastal** developments, security of vessels and mapping of coastal areas;
- Develop shared situational awareness solutions (integrating Copernicus data/services with new data sources, *e.g.*, thermal imagery, RF detections, *etc.* but also with GNSS and SATCOM) for first responders, law enforcement practitioners and **critical infrastructure** managers for **emergency management and humanitarian aid** operations, as well as **security and surveillance** applications. Support the development of applications for more efficient and safer civil protection operations and preparedness against emergency scenarios, mainstreaming them for first responders use and exploiting Copernicus Contributing Missions;
- Automatization of **insurance and finance** processes with the support of EO data and development of trustable and transparent algorithms and frameworks for green financing to assess environmental footprint and impact of investments. Digitalization of **cultural heritage** monitoring activities and integration of Copernicus data with Digital Twins models;
- Development of tools and solutions for the assessment of **ESG** targets and **sustainability** risks, including standardization and creation of a regulatory framework for carbon credits market using EU space data. In addition, develop solutions for monitoring **carbon** stock inventories, specific **forest** inventories (*e.g.*, biomass, timber volume), forest health, biodiversity and habitat mapping, deforestation and illegal logging;
- Evolve the uptake and commercialization of Copernicus, fusing Copernicus data and services with data streams from smart sensors for **precision farming** applications and livestock tracking;
- Copernicus services for forecasting **energy** production, nowcasting demand and management of electricity infrastructure. Use of EO for nowcasting/forecasting of energy production/demand, including short- and long-term variables;
- Foster the digitalization of the sustainable mineral exploration and **raw material** management (in support of the EU Critical Raw Materials Act), including *e.g.*, traceability in supply chains for electric vehicles as well as multi-scale (from satellites to UAVs and close-range instruments) and multi-sensor (*e.g.*, radar, optical, thermal, *etc.*) data-fusion and integration for safer operations and tailing management.

Long-term (next MFF) R&D priorities

Given the rapidly evolving technological achievements in many domains impacting Earth Observation (and as a result Copernicus), EUSPA defines only high-level areas as long-term R&D priorities. Some of the above needs are expected to continue being priorities in the future, while EUSPA is interested to also prioritise the uptake of the Sentinel expansion missions (*e.g.*, ROSE-L, CO2M, CHIME, LSTM, *etc.*) and Sentinel Next Generation missions for developing applications in all the market segments and supporting their value-creation in the downstream, in synergy with other EU Space Programme components. The role of the industry in the data provision pillar, is foreseen to continue being an important part of some of the services, in conjunction with Sentinels, therefore leveraging at large the upcoming Copernicus Contributing Missions for certain market segments might be another area of prioritisation. On the technological side, applied AI is expected to continue being an area of development, in addition to quantum processing for EO data.

5 Horizontal elements

As Copernicus gains in maturity in the respective services, new opportunities and challenges also arise horizontally across the various services towards enhanced synergies, efficiencies, coherence, best practices, and higher-level added value products and applications beyond single services. R&I activities under the following sections are considered key enablers to foster such horizontal research cohesion across Copernicus and should be promoted systematically – where relevant - under Copernicus service evolution topics. Programmatically, there is a need to coordinate this beyond pure R&I matters as well.

5.1 Thematic Hubs

The Copernicus Thematic Hubs were established in Copernicus 2.0 to serve as a more focused and centralised access point across the various Copernicus services bringing the full and cumulative potential of Copernicus to address specific thematic applications and maximize institutional uptake in those selected areas. They should enable faster insights for a more comprehensive knowledge driven policy assessments to facilitate informed decision-making.

The broader objectives of the Copernicus Thematic Hubs are to:

- Target user communities at EU and MS level which are not yet in the main focus of one of the Copernicus Services;
- Address horizontal topics cutting across various services;
- Foster the use of digital technologies to create added-value solutions and services;
- Promote the development and competitiveness of EU industrial ecosystems;
- Support the implementation of the EU Green Deal and other flagship EU and global policy areas.

Whilst this initial development of the Thematic Hubs is done without additional resources for the first phase of implementation, it is understood that on the longer timescale the Thematic Hubs should evolve to add functionalities, tailored products and services to address the needs of the specific user communities they address. This longer-term evolution requires targeted research and development activities to advance the state of the art. Some areas identified as needing attention to ensure this efficient evolution includes:

- The need for more resources to build dedicated products at the interface of the core Copernicus Services; combining for example different satellites *e.g.*, Sentinel-2 and –3 for the coastal zones to create specific products;
- The integration of Copernicus with other data and providers to combine existing data to address policy needs. They should be living hubs, even linked to the media in an automatic way;
- R&I should facilitate the user journey and experience with high-level products and make it easy to use. For example the EuroGEO Action Groups could become a basis for dialogue on the Thematic Hubs;
- Research is needed to enhance service user interface aspects and the visualisation of novel, complex data types (multi-view, multi-spectral, multi-polarisation, multi-dimensional data cubes, *etc.*). This includes the most effective interfaces for less technical users *e.g.*, dashboards;

- Research will also be needed in thinking about future Thematic Hubs, in line with a Thematic Hub R&I roadmap to be further confirmed and developed, addressing new topics based on emerging policy priorities (*e.g.*, such as urban aspects, biodiversity, climate adaptation, food security, environmental compliance) which will in turn require additional research and innovation;
- Cultural Heritage is another significant priority requiring a Thematic Hub. Cultural heritage users are very diverse and new requirements are emerging from different users that need different types of data to integrate. They are not used to EO and there is a need for R&I on this topic;
- Many of these topics for Thematic Hubs require integration of data from different services, data infrastructure and data governance issues will obviously also require new R&I topics.

5.2 Computing and data infrastructures

Copernicus builds on a constellation of satellites that makes a huge number of daily observations - taking advantage of a global network of thousands of land, air, and marine-based sensors to create the most detailed pictures of Earth. The technological prowess of Copernicus, especially in terms of availability and accessibility, has made Copernicus the largest space data provider in the world, currently generating 25 terabytes per day. The vast majority of data/information delivered by Copernicus is made available and accessible to any citizen, and any organisation around the world on a free, full, and open basis.

The data and information delivered by the Copernicus Service are made available to users through the websites of the service in question. In most cases, data and information can be browsed/discovered without prior registration, but registration is always required for downloading.

The Copernicus data dissemination elements was initially implemented separately for EO and respective services organic data infrastructures, largely based on developments in precursor research projects, but it was soon realized that much more could be achieved by fostering interoperability and interfacing under a dedicated framework. To facilitate and standardise access to data, the European Commission had funded the deployment of cloud-based platforms known as the DIAS, or Data and Information Access Services, to provide centralised access to Copernicus data and information, as well as to processing tools. This has recently evolved with an additional level of integration of Earth observation, and in the future with service data, through the Copernicus Data Space Ecosystem.

Similarly, the Cloud and HPC infrastructures were -and are still largely – implemented separately within each service but need to scale to the ever-increasing user requirements in terms of spatio-temporal resolution and to the full data archives and volumes with necessary processing, mining, analytics, fusion and AI-readiness capabilities to contribute to the development of new and enhanced Copernicus products and services. Big Data processing and management capabilities, and the integration with distributed data sources from other domains shall enable the development of new operational time and energy to solution capabilities, and technologies to improve relevance, performance and offer of the Copernicus value chain.

Research will be required to enhance:

- associated data governance, protocols, standards, formats, metadata, documentation, discoverability, access, dissemination according to Copernicus policies and best practices;
- interoperability, federation, querying, Linked Open Data paradigms, interactivity across platforms;

- traceability, integrity, certification, authentication, encryption, to ensure trustworthiness of products;
- system and service efficiency, quality of service, compression, scalability, latency, queuing protocols, time and energy to solution to meet growing user expectations;
- data curation, readiness, mining, sub-setting, analytics and fusion capabilities, semantics and ontologies, processing libraries and toolboxes, tutorial, trainings, and capacity building;
- level of integration and federation across data and HPC/cloud/edge infrastructures to ensure computing resources are close enough to the data so as to enable big data analytics and broader value-adding of Copernicus.

Those topics are particularly crucial for products building on data from various services, due to their horizontal and multidisciplinary dimension, such as those emerging from the Thematic Hubs.

In this context, there is a critical need to coordinate very closely with and leverage more actively EU research infrastructures and other data streams (*e.g.*, EOSC, DestinE Data Lake, WeKEO, European Weather Cloud) but also citizen science and web data at large.

Additionally, this will provide opportunities to consolidate Earth system and spatio-temporal seamless approaches and to enhance the consistency of Copernicus data, products, applications across services (*e.g.* geophysical consistency across climate data records, reanalyses, forecast products, *etc.*).

5.3 Digital Twins

A Digital Twin of the Earth (DTE) is an open and interactive system that fully integrates observations, Earth system models and data from human systems to address challenges such as climate change adaptation and mitigation, or extreme events and other natural hazards, emergency management.

Destination Earth (DestinE) is a flagship initiative of the European Commission (DG CNECT) to develop highly accurate DTE on a global scale. These models will monitor, simulate and predict the interaction between natural phenomena and human activities at a level that represents a real breakthrough in terms of accuracy, local detail, access-to-information speed and interactivity. This initiative is supported by dedicated data infrastructures and core user platform elements to address big data and interactivity challenges.

ECMWF is creating the software and data environment needed to power the high-resolution digital twins, the digital twin engine; and will also provide the first two digital twins. The Digital Twin on Weather-Induced and Geophysical Extremes will provide capabilities for the assessment and prediction of environmental extremes in support of risk assessment and management.

The Digital Twin on Climate Change Adaptation will support the analysis and testing of scenarios. This in turn will support sustainable development and climate adaptation and mitigation policy-making at multi-decadal timescales, at regional and national levels. Analogue efforts are developing for other digital twins, such as the Digital Twin Ocean supported by DG MARE and DG RTD or the ESA EO-base Digital Twin Earth. Those products and applications, once operational, could eventually make their way into the Copernicus services.

The concept of a digital twin of the Earth cannot be dissociated from high performance/cloud computing, big data, physics-based models and data assimilation. Moreover, the development of interactive Digital

Twins represents a clear opportunity to transform the Copernicus portfolio, workflows, data production (user-pull/on-demand vs provider push) and user interfacing.

Machine learning (and particularly deep learning) emulation of digital twins context offers the potential of reaching high dimensionality, complexity, nonlinearity and interactivity of real-life Earth systems and would expanding the learning ability from Big Data.

Progress in physics-constrained learning, causal inference and reinforcement learning will increase the interpretability of Big Data and the ability of simulations to solve sequential decision.

Future research must focus on each HPC/Cloud/Edged computing, big data, Earth system models, and deep learning methodologies and the optimal way to integrate them to achieve the full potential of interactive digital twins and transform Copernicus. Close collaboration with DG CNECT is paramount to achieve this.

5.4 Artificial Intelligence/Machine Learning

Whilst Artificial Intelligence (AI)/Machine Learning (ML) is not a new topic, it is only over the last decade or so that the simultaneous advent of big data, HPC/cloud/edge computing and new AI algorithms (*e.g.*, deep learning and generative methods) are revolutionizing almost every sector of society. Most, if not all, Copernicus implementing entities have developed specific strategies and/or activities to tap into those opportunities in support of their value chain (*e.g.* [ECMWF AI Roadmap for the next 10 years](#), Copernicus Marine Service Evolution Strategy: R&D priorities; EUMETSAT AI and Machine Learning Roadmap: driving Innovation as a Community, ESA's AI4EO challenges of the Φ -lab).

The potential is huge for AI/ML to transform the way Copernicus operates, enhance the user experience and improve the Copernicus offer, quality and timeliness of its products and services. This is an opportunity Copernicus cannot miss and requires a strong coordination with DG CNECT and its R&I planning.

AI-related research is specifically needed on:

- onboard processing to enhance detection of features of interest, speed up workflows and reduce data downlinks for focused applications;
- fast, reliable, consistent, and as much as possible sensor agnostic identification of clouds and shadows in optical sensing;
- AI-supported retrieval algorithms on both passive and active sensing for existing and upcoming Sentinel missions;
- physics parameterization and parameter optimization to emulate poorly understood processes and increase model fidelity;
- fault and outlier detection in production and delivery workflows to ensure more robust services;
- support to automated pre-processing and QA/QC of observations and data to reduce the risk of man-made errors and product deficiencies;
- data fusion techniques towards added-value products;
- data compression and mining methods to navigate big data efficiently, as the amount of data is becoming a limiting factor;

- hybrid ensemble data assimilation techniques, error calibration and uncertainty quantification towards improved (re-)analyses and forecast skill;
- analysis-driven Earth system deep learning models to boost prediction skill and timeliness, including with Digital Twin Earth models. These methods have shown great promises when applied to reanalyses for example;
- exploring the potential of large pre-trained foundation models and transfer learning at scale for Earth system modeling;
- experimenting observation(-only)-driven forecasting to support time-critical service elements, circumventing analysis steps. These approaches could be particularly suited for observation-dense areas from which processes can be inferred from observations alone;
- downscaling and super resolution applications building on Copernicus data to refine products in space and time;
- adaptive workflow optimization;
- enhanced interactive interfaces enabling on-demand product and service generation;
- chatbots that can guide the user across a wide range of information sources within and across Copernicus services for enhanced user support and experience.

Machine learning applications will have to interact with all important Copernicus and other data. Deep learning (emulated) models can replace entire modelling workflows enabling also the full activation of the interactive digital twin capabilities at unprecedented time and energy to solution, and the possibility to work with dynamic datapoints in space and time. Large pre-trained foundation machine learning models offer the potential for transfer learning at scale into wider Earth system applications.

Given the QA/QC requirements on Copernicus products, explainable, trustworthy and open-source AI approaches are of particular interest, as AI mainly operates as a black box. A robust framework is required to ensure the same stringent quality, reliability, and verifiability requirements of AI-generated products, as well as transparency and clearly labelled information to users. This is particularly true for generative AI and natural language processing.

It is recommended to pursue three main lines of R&I activities towards disruptive capabilities for Copernicus in the near future: 1) data curation and AI-ready data; 2) modelling, analytics, mining, fusion, and prediction; and 3) interface, visualization and user experience (cf generative AI). This will require a simultaneous coordination of R&D activities on the three broader key AI/ML ingredients (big data, computing and algorithms). Here again, a close collaboration with DG CNECT is paramount to achieve this ambition.

5.5 Synergies and partnerships at European and Global level

Close cooperation between DG DEFIS and DG RTD is more important than ever to ensure Copernicus stays at the forefront, benefits from the latest science and innovation and feeds back user requirements to research.

The previous chapter outlined the numerous R&D synergies already existing at each entity level to ensure proper implementation of their entrusted tasks. The expanding pan-Copernicus initiatives such as the Thematic Hubs adding value beyond single services highlight the need for R&I synergies and partnerships across corresponding Copernicus implementing entities.

Copernicus builds intrinsically also on Member States' R&I initiatives and expertise which helps identifying gaps and answering operational needs. R&I common to a critical mass of Member States not covered by the current Copernicus offer could be addressed through a dedicated multinational partnership element in the programme for example, subject to programmatic decisions. One shall note that the EU also offers financial instruments, such as INTEREG, for such initiatives.

At the Commission level, the same need for synergies and partnerships holds across relevant European Commission Directorate Generals to inform on policy needs and subsequent research requirements.

Synergies with DG CNECT and Destination Earth are particularly crucial to leverage the latest development in digital infrastructures, digital twins and AI/ML, which offer great potential to transform the Copernicus portfolio and user experience towards a Copernicus 3.0 digital revolution.

On the international scene, there is a need to cooperate closely at working level with international bodies such as the Committee on Earth Observation Satellites (CEOS), the Coordination Group on Meteorological Satellites (CGMS), the European Association of Remote Sensing Companies (EARSC), the Group on Earth Observation (GEO), the Open Geospatial Consortium (OGC), *etc.*

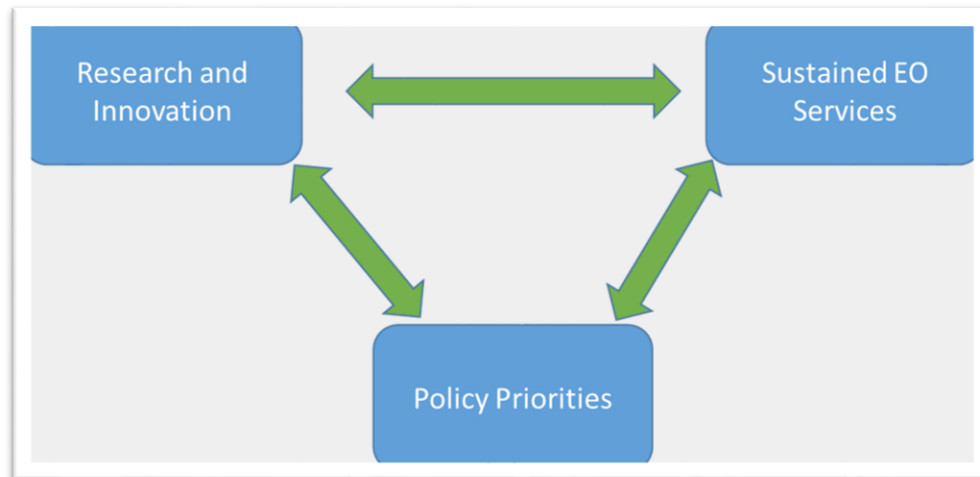
EuroGEO, the European regional GEO initiative delivers downstream services for end-users building on Copernicus services and other EO data. The tools and experiences of this community can also feed back into the evolution of Copernicus services. Future activities could foster this loop, and also strengthen the link with the private sector as providers of value-added services or as an end-user. An activity that supports the role of public procurers of EO services is already planned for 2024 and may need follow up.

The GEO flagships or GEO initiatives offer an opportunity for global outreach of EO based services made in Europe. For example, the European Commission is engaging in the GEO Blue Planet initiative by financing its European Office which is hosted at Mercator Ocean International (MOI), the entity also implementing the Copernicus Marine Service. GEO Blue Planet delivers actionable information to coastal communities on issues like marine litter, Eutrophication, or Sargassum in Europe and the Caribbean and supports fisherman in Africa with target marine information. This example demonstrates the opportunities of a close relationships of Copernicus Services and GEO initiatives, which should be brought forward by R&I, where possible, in the coming years. The new Thematic Hubs in Copernicus might be one supportive element to achieve this.

6 Measure of success and implementation recommendations

There are three fundamental dimensions that need to be addressed for an effective implementation of this strategy. These are reflected in the schematic below and are the Earth observation research and innovation activities themselves, the delivery and evolution of sustained Earth Observation Services and the underlying policy needs for data and information derived from Earth observation.

Figure 2. Fundamental dimensions for implementation strategies



Source: KCEO

To monitor the successful implementation of the strategy a key focus should be on the interfaces and connections between the different dimensions identified. This then allows to identify specific metrics that can be followed. These could include:

1. The number of research projects addressing emerging policy priorities, including the development of last mile policy applications building on Core Service outputs;
2. The number of research projects that address gaps identified by Entrusted Entities in their implementation of Core Services to address policy needs;
3. The number of cases where Core Services take up the output of the research project thus providing sustained policy applications, based on state-of-the-art research.

It is obvious that many of these aspects require a strong coordination between the research project consortia and the Entrusted Entities for the Copernicus Core Services. This should happen from the project inception, through implementation to review and evaluation. At the same time mechanisms implemented should account for differences across Entrusted Entities in how they can participate and engage with research consortia, as well as the scalability in resources to follow numerous research projects at the same time.

Competition under those activities should be open, but Entrusted Entities should also be eligible and equally encouraged to apply and/or get involved in proposals as much as Horizon Europe permits. In particular, projects shall include dedicated efforts to develop transition pathways for subsequent uptake in the services.

Equally important is to foster the explicit exploitation of Copernicus data, products and applications in R&D activities across funding instruments, *e.g.* via eligibility criteria when appropriate, in close cooperation with implementing EU executive agencies to further inform the programme evolution.

Furthermore, a cross-cutting aspect of this strategy, which will need to be monitored, is the ambition to mainstream Europe's Earth Observation research throughout the Framework Programme and beyond. This would ensure the complementarity of different calls to address identified needs and Service evolution priorities and should include all of the DGs and Executive Agencies having responsibility for the different research areas. A specific suggestion would be to organise dedicated discussions across these stakeholders at appropriate points in the co-creation process to ensure this complementarity, avoid overlaps and updates when appropriate. This could be organised in the context of the KCEO Working Group on Research and Innovation.

Finally, the Implementation of the strategy with respect to the three dimensions identified above also has a cautionary aspect which needs to be considered. It is apparent that both the research efforts and the sustained EO Services have interfaces to the EU policy priorities. Increasingly research projects are required to make these linkages and this is, of course, a core part of the mandate of Copernicus Services. We need to ensure that these diverse interfaces to the policy communities (at EU and national level) do not create confusion or "request saturation". This engagement with the user stakeholder communities should be a key element of the closer engagement of Entrusted Entities in research projects, as discussed above. This will ensure a seamless transition pathways of research outcomes into operations and where relevant European research infrastructure could also offer dual use. Funding should avoid dispersion of resources, *e.g.* by supporting single successful proposals per topics.

7 Conclusions

The present Earth Observation Strategic Research and Innovation Agenda (SRIA) serves as a comprehensive framework designed to guide the evolution of Copernicus. It was developed to meet growing user expectations, leverage technological innovations, and align with evolving EU and global policies.

Research and innovation are inherently in continuous evolution and progress, and the field of Earth observation is no exception to this. The current Phase II of Copernicus, the Earth observation component of the EU Space Programme, is well underway and the enhanced continuity of the services already delivers new and improved data, products, and services to their users, leveraging the latest science and technology developments.

The SRIA updates the 2020 roadmap for the current Multiannual Financial Framework (MFF) period, incorporating feedback from growing user requirements, new technologies, evolving EU and global policies, and changes in the global geopolitical context. The document will serve as a reference for calls issued for the period of 2025-2027 and sets broad directions for the next MFF, with plans for further updates before 2028.

The agenda emphasizes the need for agile and resilient Earth observations, service enhancements through digital transformation, the development of federated data infrastructures, and the promotion of the downstream market uptake and mass adoption of Copernicus products. The main initial tenets of the Copernicus evolution remain largely valid. Yet the SRIA also highlights the need for an agile approach to R&I for Earth observation and Copernicus.

The SRIA aims to inform long-term research and innovation, justify investments, and guide the development of research calls over the years. It was developed in coordination with various EU entities and builds on extensive consultations with stakeholders, including Entrusted Entities, Executive Agencies, Member States, to mainstream research and innovation across Horizon Europe.

The document highlights both specific and broader research priorities for the Copernicus services, including climate change, atmosphere monitoring, emergency management, land monitoring, marine environment monitoring, and security services.

The report also discusses the importance of upstream space and ground segments, and downstream elements such as market uptake, and horizontal elements like thematic hubs, computing and data infrastructures, digital twins, AI/machine learning, and synergies and partnerships at European and global levels. The need for close cooperation between various EU entities and international bodies is underscored to ensure a cohesive and effective approach to Earth observation R&I. The report proposes measures of success and makes recommendations for R&I implementation in the future.

The SRIA is a dynamic tool and shall be considered as a living document. A further update is warranted before the start of the next MFF, to ensure that a new full R&I roadmap can sustain the Copernicus evolution and ensure all services remain at the forefront to serve our core users and stimulate wide market uptake.

References

European Commission, Directorate-General for Research and Innovation, *Horizon Europe strategic plan 2025-2027 analysis*, Publications Office of the European Union, 2023, <https://data.europa.eu/doi/10.2777/637816>

European Commission, Joint Research Centre, Kristopaitis, E., Meneses De Abreu, R., *The strategic research agenda for Copernicus security service 2023 – Guidelines for Horizon Europe 2024*, Publications Office of the European Union, 2023, <https://data.europa.eu/doi/10.2760/38747>

European Court of Auditors, *EU space programmes Galileo and Copernicus – Services launched, but the uptake needs a further boost. Special report No 07*, 2021, Publications Office, 2021, <https://data.europa.eu/doi/10.2865/073967>

List of figures

Figure 1. European Green Deal Legislation including Copernicus by policy area over time	11
Figure 2. Fundamental dimensions for implementation strategies.....	58

Getting in touch with the EU

In person

All over the European Union there are hundreds of Europe Direct centres. You can find the address of the centre nearest you online (european-union.europa.eu/contact-eu/meet-us_en).

On the phone or in writing

Europe Direct is a service that answers your questions about the European Union. You can contact this service:

- by freephone: 00 800 6 7 8 9 10 11 (certain operators may charge for these calls),
- at the following standard number: +32 22999696,
- via the following form: european-union.europa.eu/contact-eu/write-us_en.

Finding information about the EU

Online

Information about the European Union in all the official languages of the EU is available on the Europa website (european-union.europa.eu).

EU publications

You can view or order EU publications at op.europa.eu/en/publications. Multiple copies of free publications can be obtained by contacting Europe Direct or your local documentation centre (european-union.europa.eu/contact-eu/meet-us_en).

EU law and related documents

For access to legal information from the EU, including all EU law since 1951 in all the official language versions, go to EUR-Lex (eur-lex.europa.eu).

Science for policy

The Joint Research Centre (JRC) provides independent, evidence-based knowledge and science, supporting EU policies to positively impact society



EU Science Hub

joint-research-centre.ec.europa.eu



Publications Office
of the European Union